CHAPTER 1: INTRODUCTION OF THE SUMMER TRAINING COURSE

1.1 About the Summer Training Course

The Summer Training Program on Generative Artificial Intelligence (AI) was a comprehensive skill development initiative organized by IBM during the months of June to July. The main objective of the training was to familiarize students with the practical aspects of Artificial Intelligence, Machine Learning, and Data Science, along with hands-on exposure to IBM's proprietary Watson AI platform and IBM Cloud environment. The course was thoughtfully structured to provide both theoretical knowledge and real-world applications, thereby preparing participants to understand how AI is transforming industries across sectors such as healthcare, finance, education, and automation.

The training began with setting up the programming environment by installing Anaconda, which served as a platform for managing Python libraries and running Jupyter Notebooks. Through this, participants were introduced to Jupyter Notebook, an open-source web application that enables users to create and share documents containing live code, equations, visualizations, and explanations. This formed the base for all subsequent programming exercises.

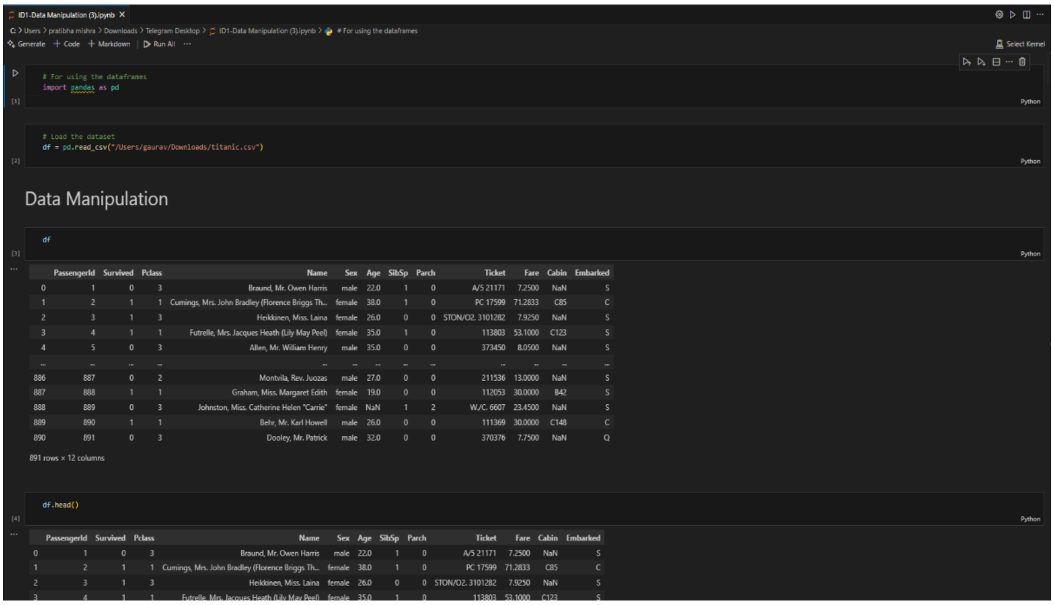

Figure 1.1.1: Data Manipulation on Jupyter Notebook.
Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM Data Science Course, 2025).

As the course progressed, participants explored data manipulation and analysis using the Pandas library. They learned how to import, clean, transform, and preprocess datasets, which is a crucial step in data science workflows. Later, the training introduced data visualization techniques using libraries such as Matplotlib and Seaborn, which helped participants understand how to convert raw data into graphical formats like bar charts, scatter plots, and heatmaps.

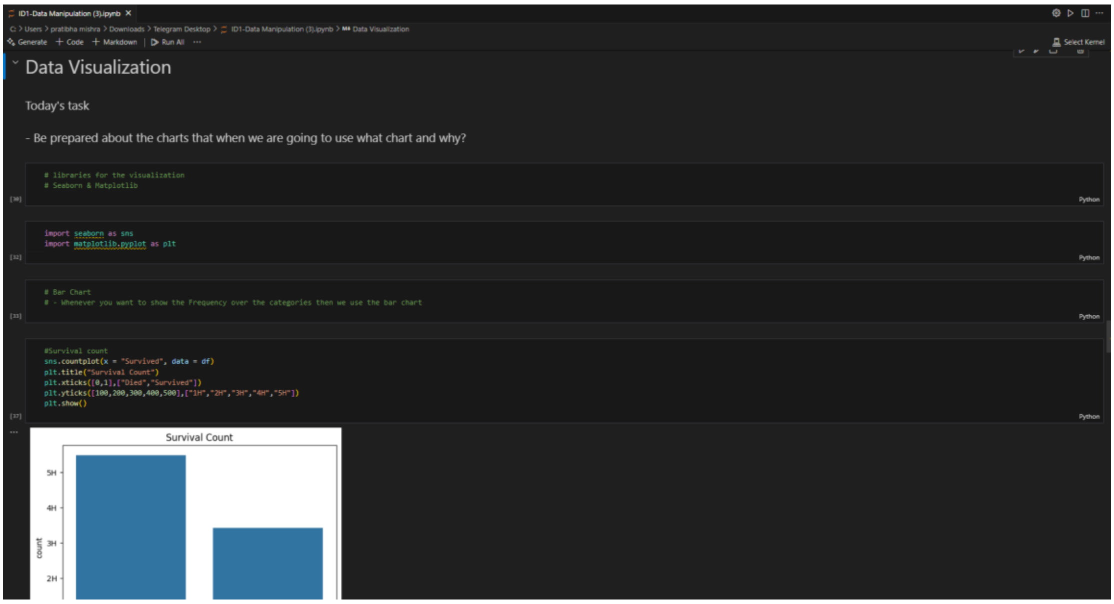

Figure 1.1.2: Data Visualisation on Jupyter Notebook.
Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM GenAI Course, 2025).

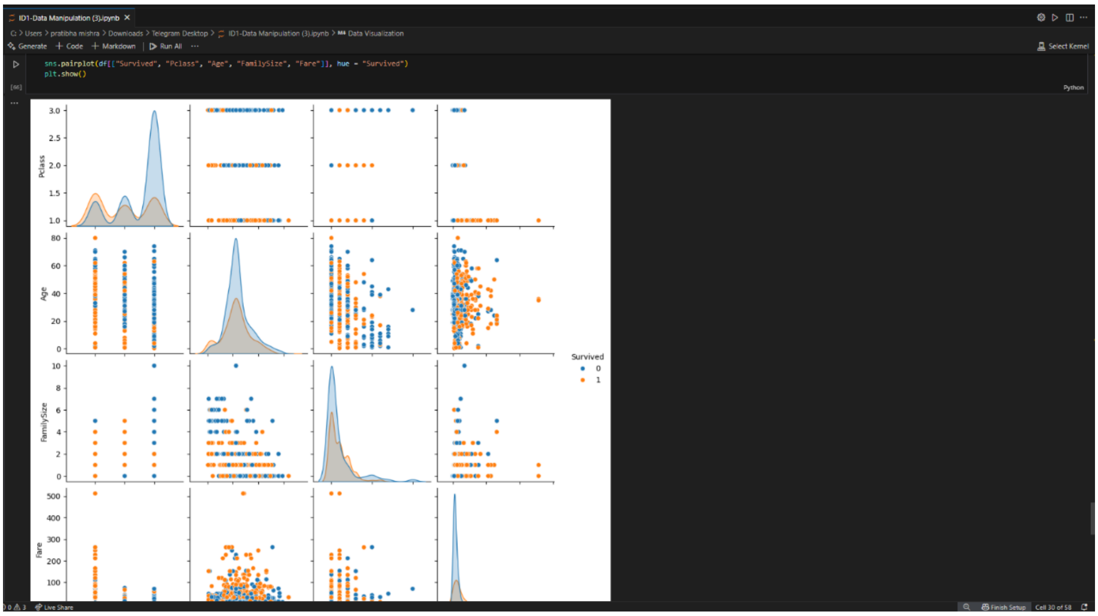

Figure 1.1.3: Graphs of Data Visualization.
Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM GenAI Course, 2025).

Once the basics were mastered, the training advanced into Machine Learning algorithms. Participants learned about Linear Regression and Logistic Regression, which are the foundation of predictive modeling. Following that, advanced algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees were explored to enhance classification and prediction capabilities. The module on Ensemble Learning introduced learners to techniques like Bagging, Boosting, and Random Forests, which improve model stability and performance.

Furthermore, the course covered Unsupervised Learning algorithms such as K-Means Clustering and Hierarchical Clustering, allowing participants to identify patterns in unlabelled datasets. Later, Neural Network concepts were discussed using TensorFlow, focusing on Feedforward Propagation and model optimization techniques.

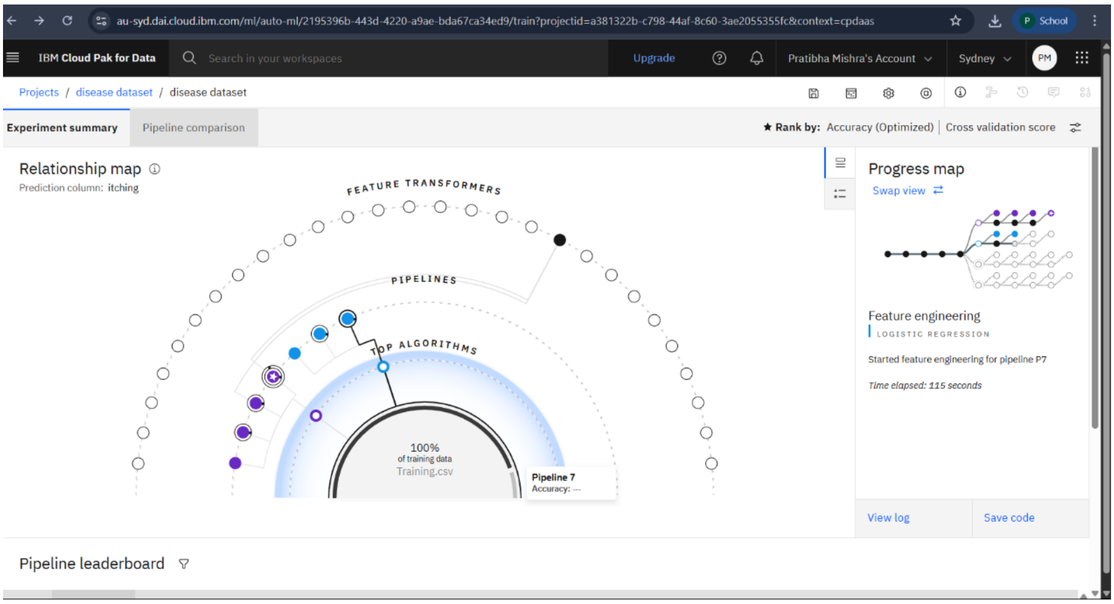

Figure 1.1.4: Analysis of Accuracy.

Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM GenAI Course, 2025).

In the final phase, the focus shifted to IBM Watson AI and IBM Cloud. Students were guided in creating their IBM Cloud accounts and introduced to Watson Studio, a platform for building and deploying AI models. They learned how Watson's AI services perform data analysis, language processing, and chatbot creation. The culmination of the training was the development of a chatbot project on "Potters", built using Watson Assistant. This project demonstrated how conversational AI can be designed, trained, and implemented for real-world scenarios. Overall, the training provided an end-to-end learning experience in modern AI techniques — from data analysis to intelligent application development.

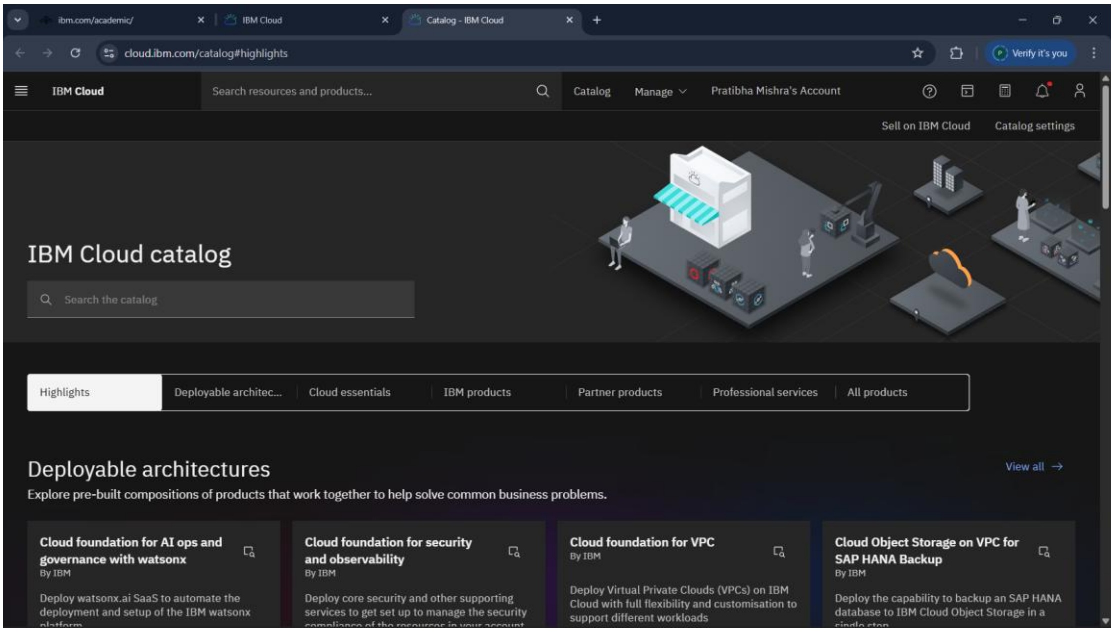

Figure 1.1.5: IBM Cloud Catalog.
Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM GenAI Course, 2025).

1.2 Profile of the Company

IBM (International Business Machines Corporation) is one of the world's oldest and most influential technology companies, headquartered in Armonk, New York, USA. Established in 1911 as the *Computing-Tabulating-Recording Company (CTR)*, it was renamed IBM in 1924. Over the years, IBM has evolved from a hardware manufacturing company into a global leader in Artificial Intelligence, Cloud Computing, Data Analytics, and Quantum Computing

Figure 1.2.1: IMB Corporation Identity.
Source: Image generated by Author using Google Gemini AI (2025).

IBM has played a pivotal role in shaping the modern computing world. Its legacy includes the development of the mainframe computer, the floppy disk, and the SQL database language. In recent decades, IBM has shifted its focus toward software, services, and artificial intelligence. One of its most revolutionary innovations is IBM Watson, an AI-powered system capable of understanding natural language, reasoning through vast datasets, and generating insights. Watson gained worldwide recognition after defeating human champions on the quiz show *Jeopardy!* — showcasing the real power of AI.

Table 1.2.1: Training Objective.

Feature	**Details/Description**	**Key Points/Impact**
Founding & History	Established in **1911** as the Computing-Tabulating-Recording Company (CTR). Renamed **IBM in 1924**.	One of the world's oldest and most influential technology companies.
Headquarters	Armonk, New York, USA.	Global presence and operational base.
Modern Focus/Core Areas	Global leader in **Artificial Intelligence (AI), Cloud Computing, Data Analytics, and Quantum Computing**.	Shifted focus from hardware to software and services.
Key Historical Legacy	Developed the **mainframe computer**, the **floppy disk**, and the **SQL database language**.	Pivotal role in shaping the modern computing world.
Revolutionary AI Innovation	**IBM Watson** – an AI-powered system capable of understanding natural language and reasoning.	Gained fame by defeating human champions on the quiz show *Jeopardy!*

Feature	Details/Description	Key Points/Impact
Global Presence	Operates in **more than 170 countries** and employs hundreds of thousands of professionals.	Massive global reach and workforce.
Major Business Divisions	Cloud & Cognitive Software, Global Business Services, Global Technology Services, Systems, and Research & Development.	Diverse structure for delivering technology solutions.
Commitment to Education	Global learning initiatives like **IBM SkillsBuild, IBM Academic Initiative,** and specialized training programs.	Empowers students/professionals with future-ready skills in AI, ML, and data science.
Guiding Vision/Motto	**"Think"**	Inspires innovation and use of technology for smarter, more efficient, and inclusive systems.

Today, IBM operates in more than 170 countries and employs hundreds of thousands of professionals worldwide. Its business divisions include Cloud & Cognitive Software, Global Business Services, Global Technology Services, Systems, and Research & Development. IBM's AI services are used by leading organizations to automate customer interactions, improve business intelligence, and streamline operations.

IBM's commitment to education is evident through its global learning initiatives like IBM Skills Build, IBM Academic Initiative, and specialized training programs such as the Generative AI Summer Training. These programs aim to empower students and professionals with future-ready skills in data science, machine learning, and cognitive computing. The company's focus on innovation, ethics in AI, and sustainability makes it a global leader not only in technology but also in shaping the responsible future of AI.

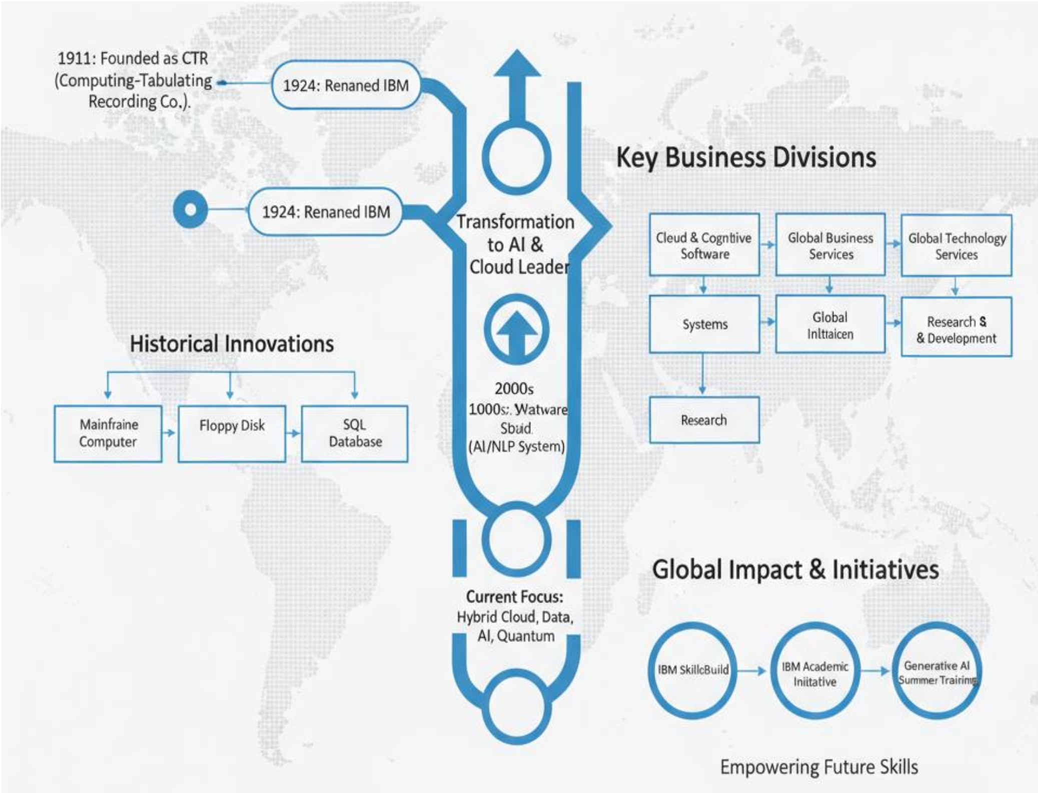

Figure 1.2.2: Evolution and Strategic Roadmap.
Source: Image generated by Author using Google Gemini AI (2025).

IBM's motto, *"Think,"* continues to symbolize its vision — to inspire innovation and to use technology for creating smarter, more efficient, and inclusive systems.

CHAPTER 2: INTRODUCTION OF THE TRAINING

2.1 About the Training

The Generative AI Training conducted by IBM was a specialized program tailored to provide students with a solid grounding in data science, machine learning, and artificial intelligence concepts. The training followed a structured, step-by-step approach — beginning with fundamental programming and data handling skills, progressing into advanced AI techniques, and finally applying those concepts in a real-world project using IBM Watson AI.

The first step of the training involved learning how to install and configure Anaconda, an open-source Python distribution that simplifies package management and deployment. Once the setup was complete, students began using Jupyter Notebook, an interactive coding environment where they practiced Python-based data analysis and visualization tasks. This hands-on approach made it easier to understand the working of code cells, outputs, and visualizations in real time.

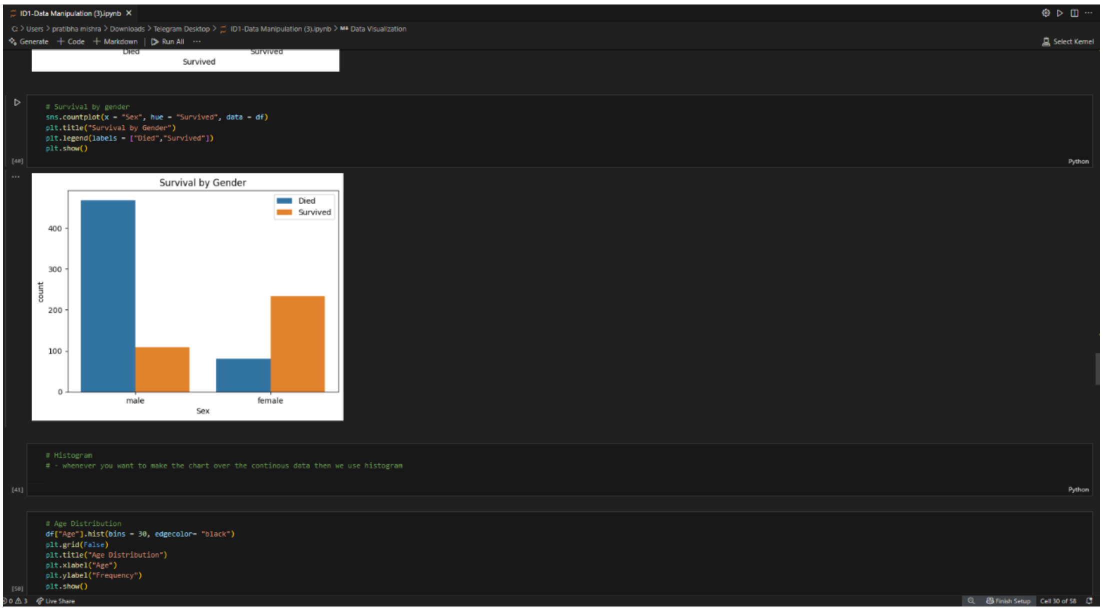

Figure 2.1.1: Graphical Representation and Data Visualisation.
Source: Screenshot captured by Author during hands-on practice in Jupyter Notebook (IBM GenAI Course, 2025).

The next segment introduced participants to the Pandas library, which is essential for data manipulation and cleaning. Learners explored how to load datasets, handle missing values, merge or split data, and perform statistical analysis. Following this, the training emphasized data visualization, an integral part of data science. Participants learned to use libraries like Matplotlib and Seaborn to generate graphical representations such as scatter plots, histograms, and heatmaps that revealed patterns within data.

Once the foundation was built, the course advanced into machine learning. Students studied supervised learning algorithms like Linear Regression for continuous predictions and Logistic Regression for binary classification. More complex algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees were covered with practical coding exercises. They also explored Ensemble Learning techniques — Bagging, Boosting, and Random Forests — that combine multiple models to improve prediction accuracy.

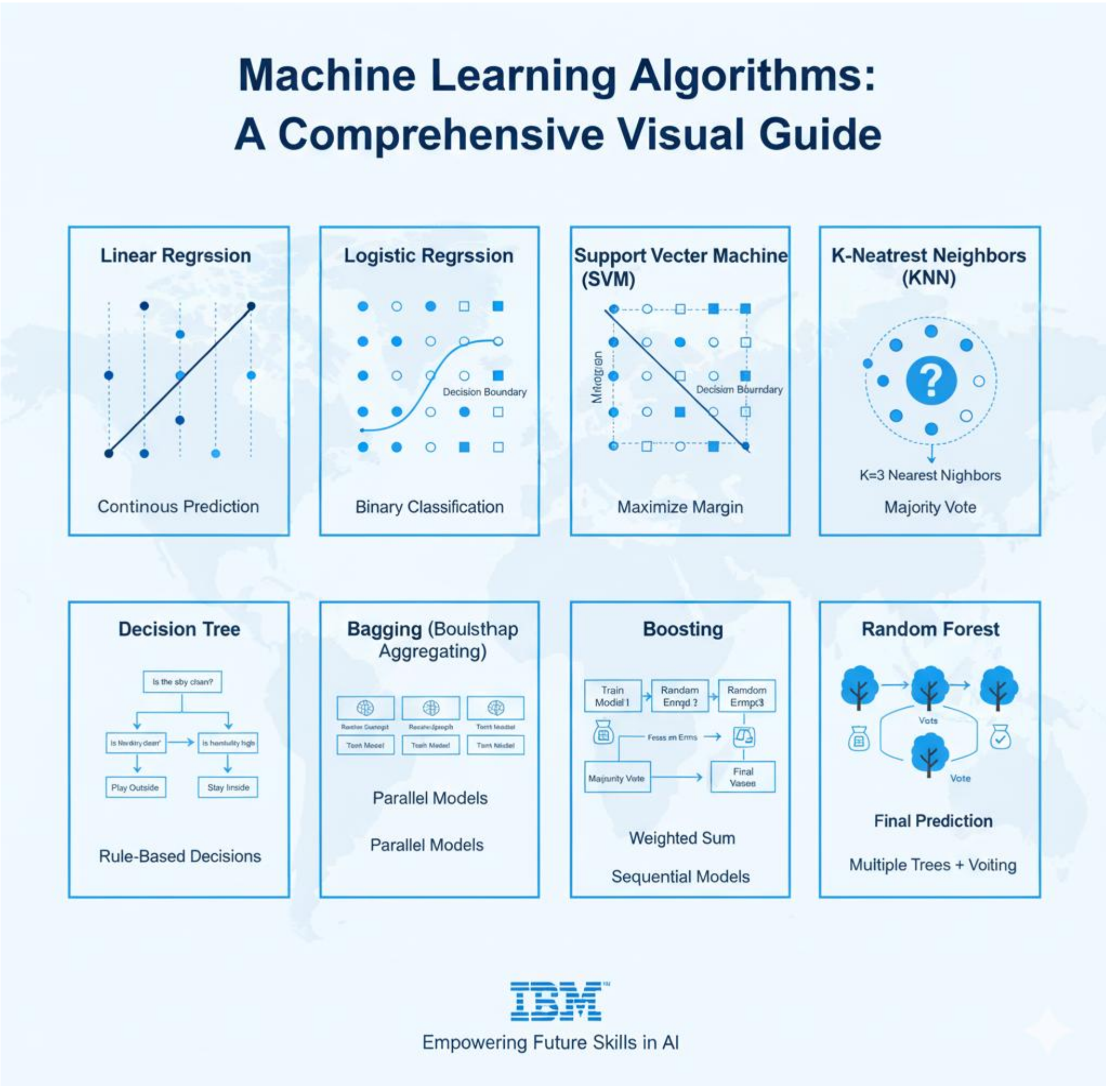

Figure 2.1.2: Machine Learning Algorithms.
Source: Image generated by Author using Google Gemini AI (2025).

Table 2.1.1: Machine Learning Algorithms.

Algorithm	One-Line Explanation
Linear Regression	A **supervised learning** model used for **continuous predictions** by fitting the best straight line through data points.
Logistic Regression	A **supervised classification** model that uses the sigmoid function to estimate the probability of a data point belonging to one of **two classes** (binary classification).
Support Vector Machine (SVM)	A **supervised classification** model that finds the optimal **hyperplane** (decision boundary) with the maximum margin between the closest data points of different classes.
K-Nearest Neighbors (KNN)	A simple, **non-parametric classification/regression** model where a new data point is classified based on the majority class of its **k closest neighbors**.
Decision Tree	A **supervised classification/regression** model that uses a **tree-like structure** of decisions (splits) based on feature values to reach a conclusion (leaf node).
Bagging	An **ensemble technique** that trains multiple independent models (e.g., Decision Trees) on **different subsets** of the training data (bootstrap samples) and averages their results to reduce variance.
Boosting	An **ensemble technique** that trains a sequence of weak models, where each new model focuses on correcting the errors (misclassified points) made by the **previous models** to improve accuracy.

Algorithm	One-Line Explanation
Random Forest	An **ensemble technique** that builds an entire **collection of uncorrelated Decision Trees** (using both data and feature subsets) and combines their predictions (by averaging or voting) to enhance accuracy and stability.

The later modules delved into unsupervised learning, focusing on K-Means Clustering and Hierarchical Clustering to identify hidden patterns within unlabeled data. The training also provided an introduction to Neural Networks and TensorFlow, where students learned about Feedforward Propagation and how computers learn from data iteratively to make predictions.

The final and most exciting part of the training revolved around IBM Watson AI and IBM Cloud. Students learned how to create accounts, navigate the IBM Cloud dashboard, and use Watson's tools for AI model development. They explored how Watson AI performs data analysis, natural language processing, and chatbot creation. The culmination of this learning journey was the development of a Potters Chatbot, a conversational AI project that integrated data-driven learning and cloud deployment. This project helped participants understand the end-to-end workflow of AI — from problem definition to implementation and testing.

2.2 Objectives of the Training

The primary objectives of the summer training were to develop technical proficiency, analytical thinking, and practical problem-solving abilities in the field of Artificial Intelligence. The key goals were:

1. To understand AI fundamentals — developing conceptual clarity about how machines learn from data and how algorithms simulate human-like intelligence.
2. To gain practical experience in Python programming and data science libraries like Pandas, NumPy, Matplotlib, and Seaborn for data manipulation and visualization.
3. To explore core machine learning algorithms including regression, classification, and clustering models, and to understand their mathematical foundations and use cases.

4. To apply ensemble learning techniques (Bagging, Boosting, Random Forests) to enhance prediction accuracy and reduce overfitting.
5. To develop working knowledge of neural networks and TensorFlow, understanding feedforward propagation and model training.
6. To understand the IBM Watson AI ecosystem — including Watson Studio, Watson Assistant, and IBM Cloud — and how they enable AI deployment at scale.
7. To gain hands-on experience in real-world AI applications, specifically through chatbot creation and dataset analysis.
8. To enhance problem-solving, teamwork, and analytical abilities, which are crucial for professional success in AI-related roles.
9. To bridge the gap between academic learning and industrial requirements, ensuring students are job-ready and technically competent in AI technologies.

Through these objectives, the training encouraged participants not just to learn coding, but to think critically, handle real datasets, and deploy intelligent solutions independently.

Tables 2.1.2: Overview of IBM.

S.No.	Primary Objective Category	Specific Key Goals & Skills Developed
1	**AI Fundamentals & Conceptual Clarity**	Develop conceptual clarity on how machines learn from data and how algorithms simulate human-like intelligence.
2	**Programming & Data Science**	Gain practical experience in **Python** programming and data science libraries (**Pandas, NumPy, Matplotlib, Seaborn**) for data manipulation and visualization.
3	**Core Machine Learning**	Explore **core ML algorithms** (regression, classification, clustering), including their mathematical foundations and use cases.

S.No.	Primary Objective Category	Specific Key Goals & Skills Developed
4	**Advanced ML Techniques**	Apply **ensemble learning** techniques (Bagging, Boosting, Random Forests) to enhance prediction accuracy and reduce overfitting.
5	**Deep Learning & Frameworks**	Develop working knowledge of **neural networks** and **TensorFlow**, understanding feedforward propagation and model training.
6	**Industry Tools & Ecosystem**	Understand the **IBM Watson AI ecosystem** (Watson Studio, Watson Assistant, IBM Cloud) for scalable AI deployment.
7	**Hands-on Application**	Gain hands-on experience in **real-world AI applications** through chatbot creation and dataset analysis.
8	**Soft Skills & Professional Readiness**	Enhance **problem-solving, teamwork, and analytical abilities** crucial for professional success in AI roles.
9	**Academic-Industry Bridge**	**Bridge the gap** between academic learning and industrial requirements to ensure students are job-ready and technically competent.

2.3 Roles and Responsibilities

Throughout the training, students were encouraged to take ownership of their learning process by performing specific roles and tasks assigned by the instructors. These responsibilities helped in reinforcing both technical and soft skills.

- Active Engagement: Students were expected to attend every session, interact with trainers, and complete practical assignments within the given deadlines.

- Software Setup and Configuration**:** Each participant was responsible for installing Anaconda, setting up the Python environment, and managing Jupyter Notebook files efficiently.
- Data Handling and Preparation: Participants learned to import, clean, and preprocess datasets using Pandas. This included handling missing values, normalizing data, and preparing it for machine learning models.
- Model Implementation**:** Students implemented algorithms such as Linear and Logistic Regression, SVM, KNN, and Decision Trees using real datasets to understand prediction, accuracy, and error metrics.
- Visualization and Reporting**:** Learners created visual summaries of their findings using Matplotlib and Seaborn, documenting each experiment and output.
- Collaborative Learning**:** The training encouraged teamwork by assigning group exercises, where students collaborated to troubleshoot errors, compare results, and improve model performance.
- Exploring IBM Tools**:** Each student had to create an IBM Cloud account, log into Watson AI, and understand its dashboard, services, and API functionalities.
- Chatbot Development**:** One of the major roles was to design and develop the Potters Chatbot using Watson Assistant — defining intents, entities, and dialog flow for smooth user interaction.
- Documentation and Presentation**:** Participants were responsible for compiling their weekly progress, observations, and final project reports. The course concluded with a presentation to IBM mentors, showcasing the knowledge gained during the training.

These roles instilled a sense of responsibility, self-learning, and teamwork among the participants. By the end of the training, every student had hands-on experience in developing AI applications and a solid understanding of how data-driven technologies function in the real world.

CHAPTER 3: PROBLEM STATEMENT

3.1 Software Requirement Specifications (SRS)

The Porter Chatbot – The Porter Who Delivers is an AI-powered conversational assistant developed as part of the IBM Generative AI Summer Training Program. It was designed to simplify and automate the logistics and delivery experience for customers by providing instant responses to common queries such as delivery booking, tracking, complaint registration, feedback, and wallet management.

The chatbot integrates Natural Language Processing (NLP) using IBM Watson Assistant, hosted on IBM Cloud, to simulate a human-like customer service experience. Through this project, students explored practical applications of Generative AI, machine learning, and automation to create a realistic and user-friendly virtual support system.

The chatbot acts as a digital delivery assistant, available 24×7, to reduce human workload and enhance customer satisfaction by offering accurate, personalized, and instant responses.

3.1.1 Functional Requirements of the Project

Functional requirements define the core operations that the system performs. For the Porter Chatbot, the following functional requirements are based on its five major functional areas and core capabilities.

1. Delivery Assistance

Description:

The chatbot helps users schedule or inquire about delivery services.

Functions include:

- Accept user requests for pickup and delivery.
- Ask for delivery details (pickup location, destination, parcel type, and preferred delivery time).
- Generate a delivery summary for user confirmation.
- Provide fare estimates based on weight, distance, and urgency.
- Forward delivery details to the system database for processing.

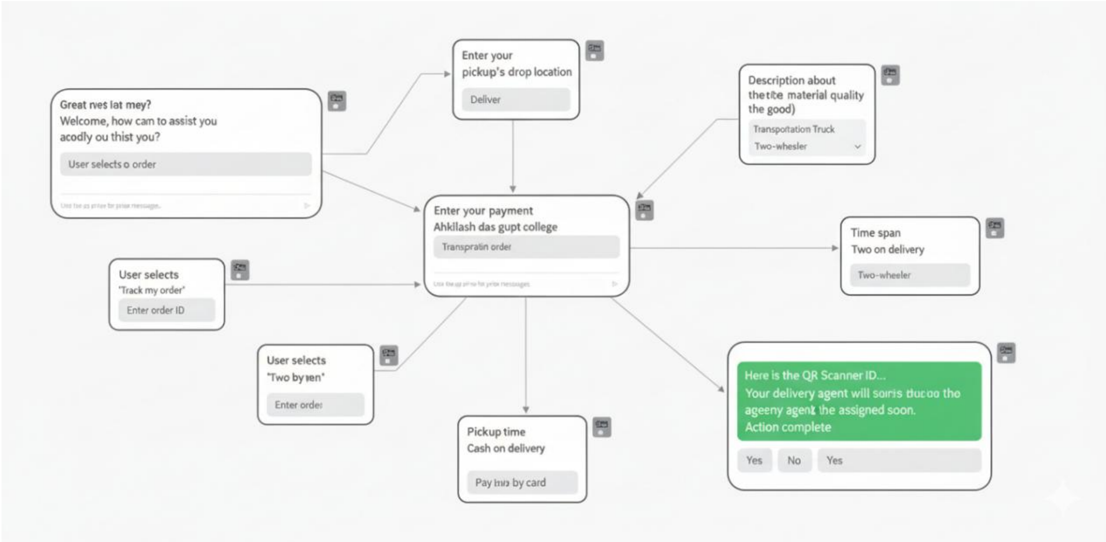

Figure 3.1.1.1.1(a): Delivery Assistance Interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

Expected Output: A clear delivery confirmation message with delivery ID, fare, and estimated time.

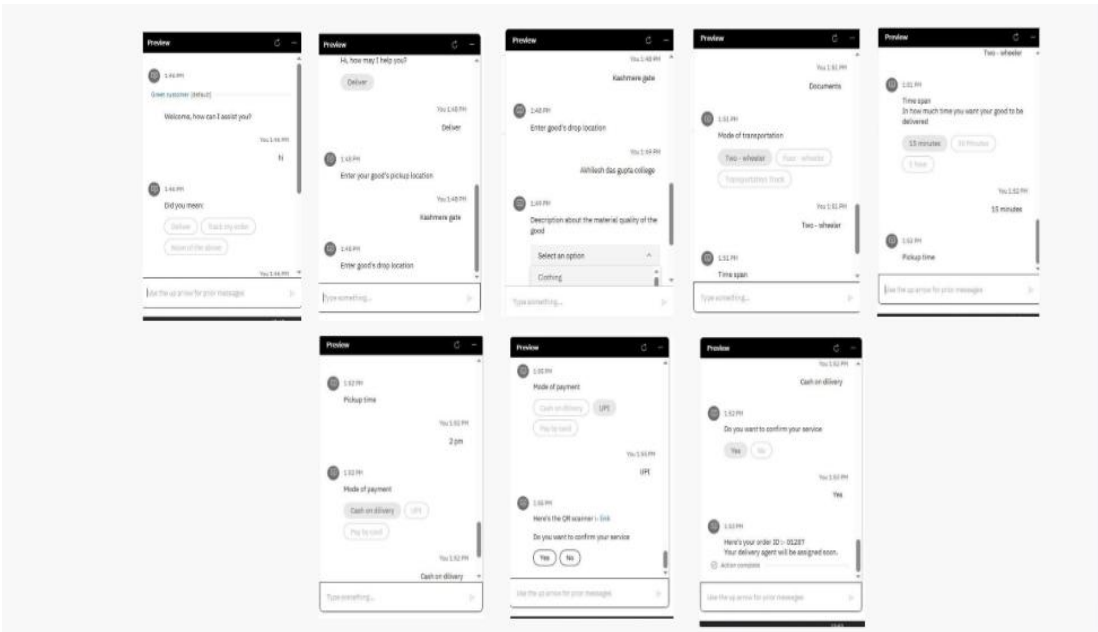

Figure 3.1.1.1.2(b): Delivery Assistance Screen.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025)

2. Track My Order

Description:

Allows customers to check the status and location of their parcels in real time.

Functions include:

- Accept tracking number or registered mobile number.
- Retrieve delivery data from the backend.
- Display current status (e.g., “In Transit,” “Out for Delivery,” “Delivered”).
- Provide estimated delivery time and courier contact information.

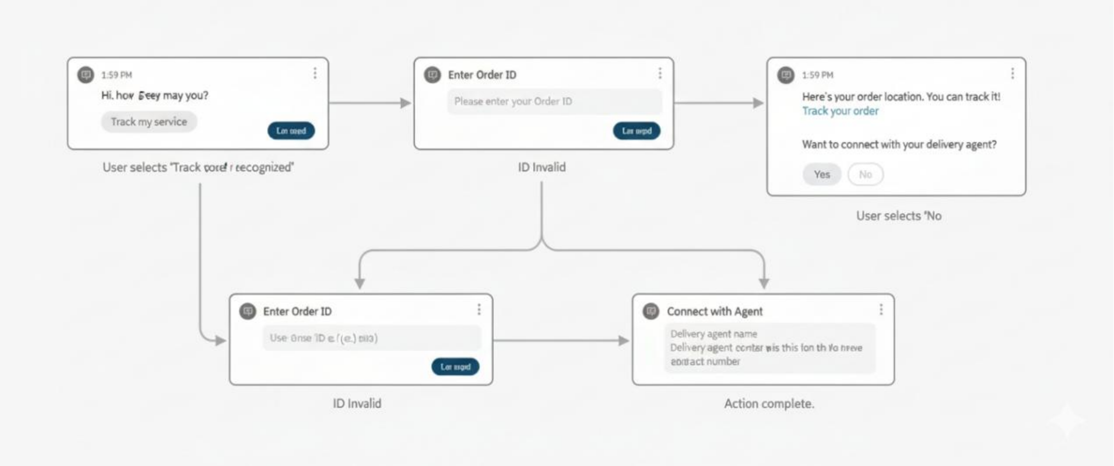

Figure 3.1.1.1.3(a): Oder Tracking interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

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Expected Output:

Instant tracking update with real-time order status and timestamp.

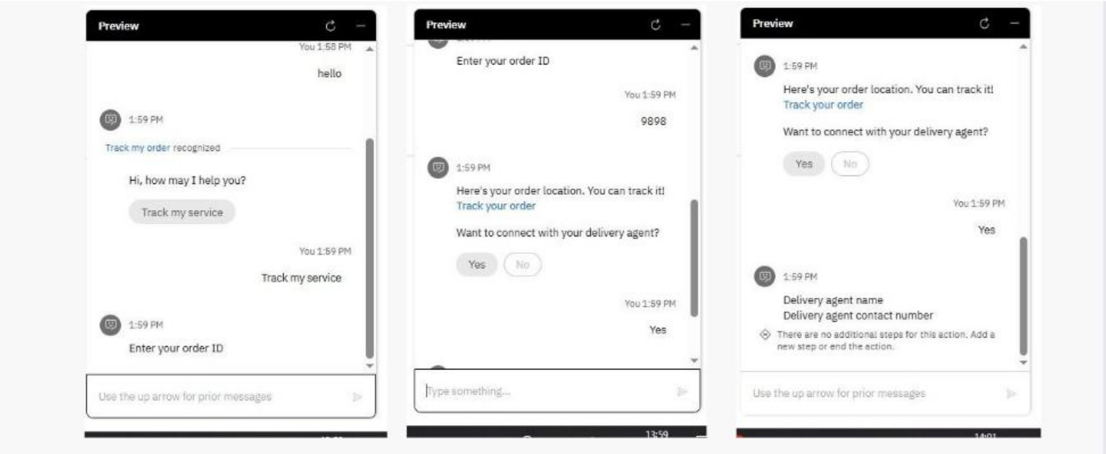

Figure 3.1.1.1.4(b).: Oder Tracking Screens.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025).

3. Feedback Collection

Description:

The chatbot collects user feedback after service completion to help improve quality and customer experience.

Functions include:

- Prompt users for service feedback after delivery completion.
- Record comments, ratings, or suggestions.
- Store feedback data in the system for analysis by the admin team.
- Provide a thank-you acknowledgment to encourage customer participation.

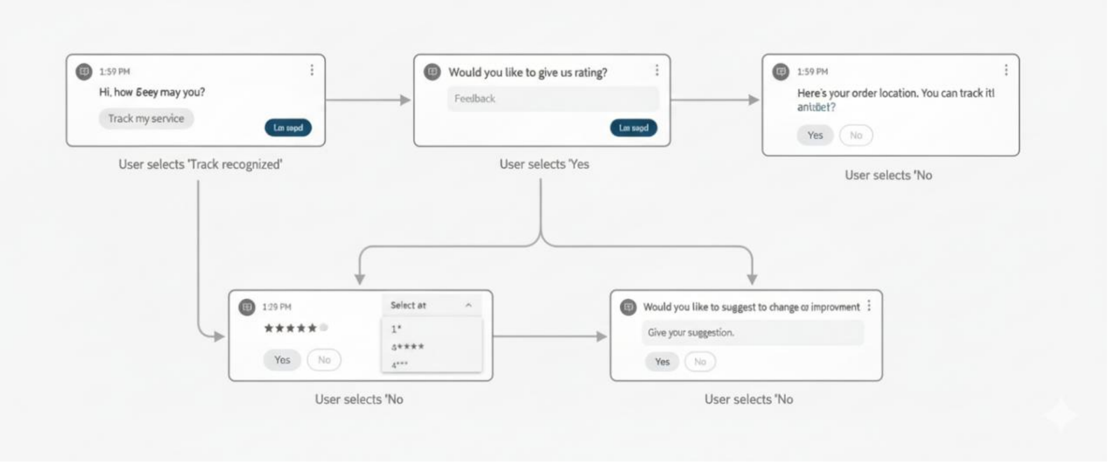

Figure 3.1.1.1.5(a): Feedback Collection Interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

Expected Output:

Feedback successfully stored and confirmation message displayed.

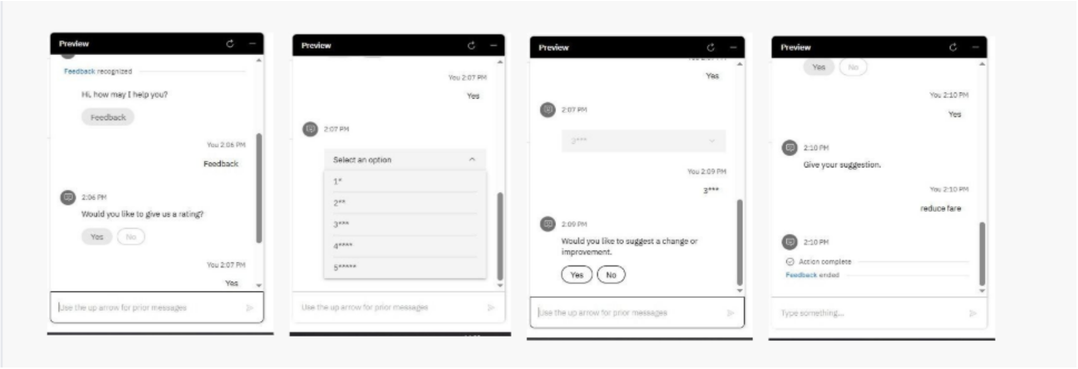

Figure 3.1.1.1.6(b): Feedback Collection Screens.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025).

4. Complaint Handling

Description:

Enables users to report problems such as delayed delivery, lost parcels, or unsatisfactory service.

Functions include:

- Accept complaints through natural conversation.
- Classify complaint type (delay, loss, behavior issue, etc.) using intent recognition.
- Generate a complaint ID for user reference.
- Forward complaint details to the admin or support team automatically.

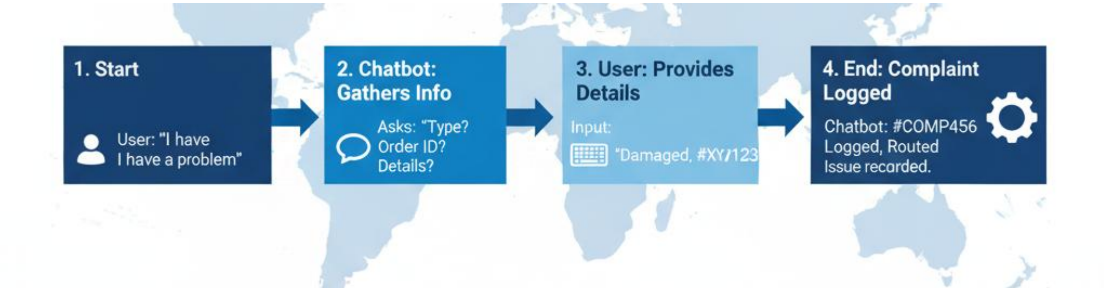

Figure 3.1.1.1.7(a): Complaint Handling Interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

Expected Output: Complaint ID with an assurance message: “Your complaint has been registered. Our team will contact you soon.”

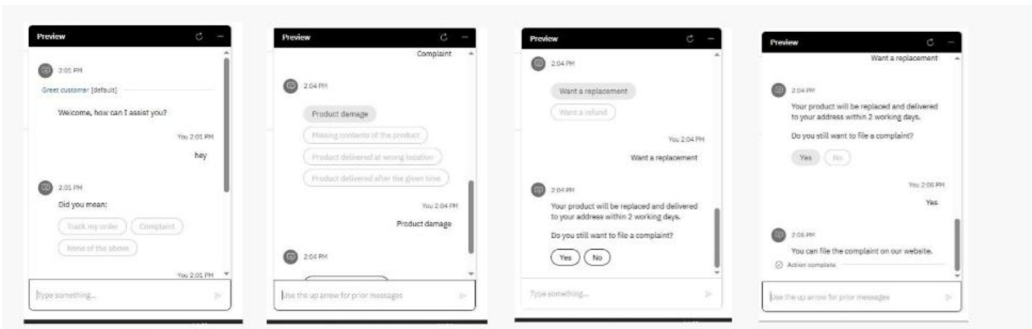

Figure 3.1.1.1.8(b): Complaint Handling Screens.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025)

5. Wallet Points Management

Description:

Manages user reward points and transactions within the chatbot ecosystem.

Functions include:

- Display available wallet balance or reward points.
- Update wallet points after successful deliveries or referrals.
- Notify users about redemption options.
- Maintain transaction history securely.

Figure 3.1.1.1.9(a): Wallet Point Interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

Expected Output: Wallet summary showing total points, transaction details, and redemption options.

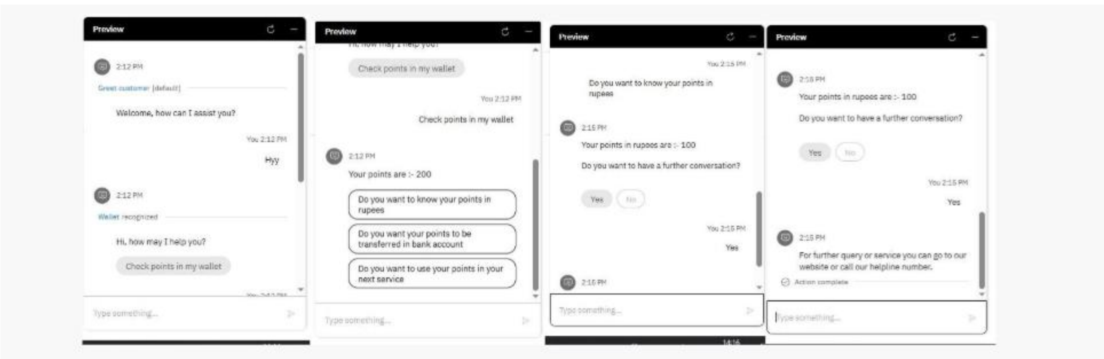

Figure 3.1.1.1.10(b): Wallet Point Screens.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025).

6. Rating System

Description:

Allows users to rate the delivery experience on a 1–5 scale.

Functions include:

- Prompt for rating after delivery confirmation.
- Store ratings for each delivery ID.
- Display thank-you or follow-up message.

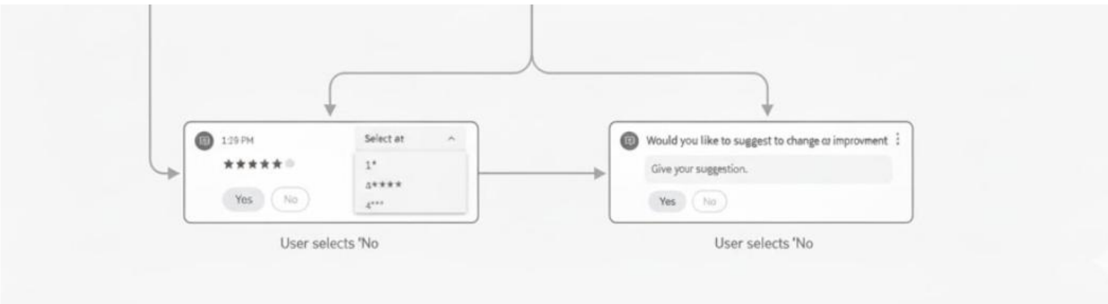

Figure 3.1.1.1.11(a): Rating System Interaction Flow.
Source: Image generated by Author using Google Gemini AI (2025).

Expected Output:Rating saved and average rating displayed to the user.

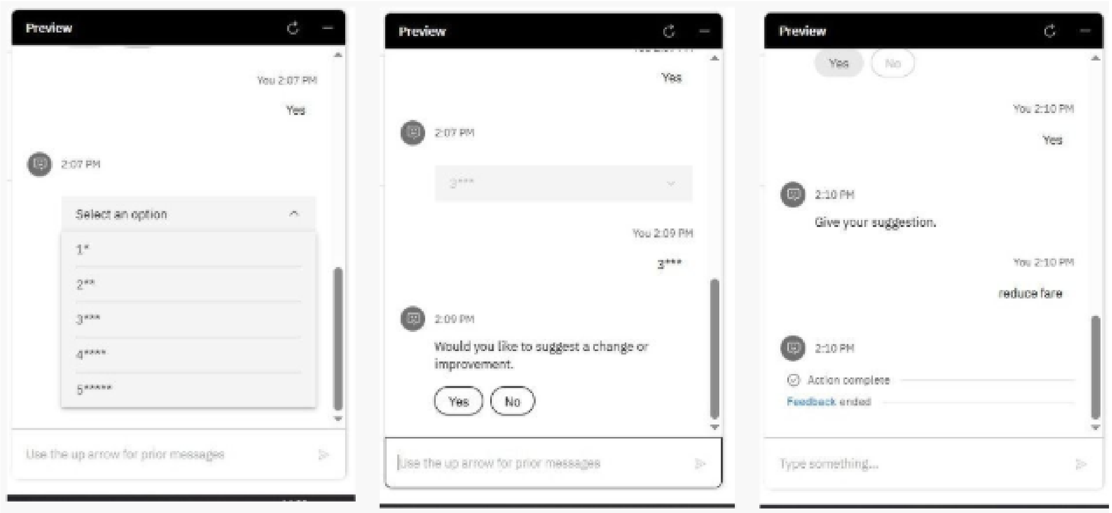

Figure 3.1.1.1.12(b): Rating System Screens.
Source: Screenshot captured by Author during chatbot development using IBM Watson Assistant (IBM GenAI Course, 2025).

3.1.2 Non–Functional Requirements of the Project

Non-functional requirements describe how well the system performs its functions, ensuring reliability, usability, and scalability.

1. Performance
   - The chatbot should respond to user queries within 2–3 seconds.
   - It must handle at least 20 concurrent users smoothly.
   - Response generation should remain consistent regardless of input type.
2. Usability
   - Interface should be simple, interactive, and visually clear.
   - Users should be able to perform all actions through text input without needing technical knowledge.
   - Clear prompts and confirmations should guide user interaction.
3. Reliability
   - The chatbot should remain functional 24/7 with minimal downtime.
   - In case of technical issues, fallback messages (like *"Please try again later"*) should be triggered automatically.
4. Security
   - All personal details (names, numbers, addresses) must be securely encrypted using IBM Cloud's built-in security protocols.
   - Only authorized admin users should access stored complaint and feedback data.
5. Scalability
   - System must support future upgrades such as voice integration, multilingual support, or payment gateway integration.
   - Should allow easy addition of new intents or service features in Watson Assistant.
6. Maintainability
   - Developers should be able to update intents, entities, or dialog nodes without affecting existing functionality.
   - Logs should be available for tracking issues or performance metrics.
7. Portability
   - The chatbot should be deployable on multiple platforms — website, WhatsApp, or mobile app — with minimal configuration.

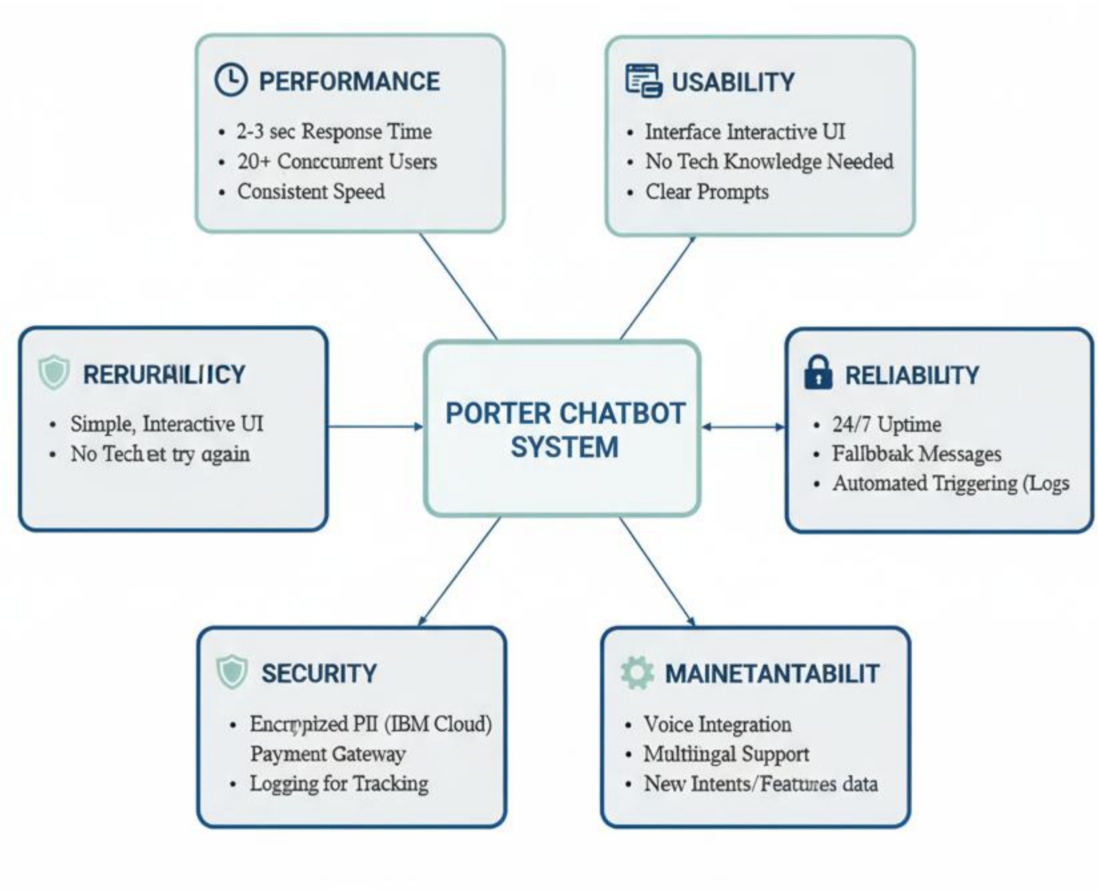

Figure 3.1.2.1: Non- Functional Requirements
. Source: Image generated by Author using Google Gemini AI (2025).

3.2 Feasibility Study of the Project

Feasibility study ensures that the project is practical, cost-effective, and technically achievable.

1. Technical Feasibility

The project was implemented using IBM Watson Assistant, which provides ready-to-use AI models for chatbot design.

Watson's drag-and-drop interface, intent recognition, and cloud deployment made it easy to create and train the chatbot.

Students used Python, Anaconda, and Jupyter Notebook for initial model understanding and IBM Cloud for final deployment.

Hence, the project is technically feasible with minimal hardware (8 GB RAM, internet access) and no specialized infrastructure.

2. Operational Feasibility

The chatbot directly supports logistics operations by automating:

- Delivery assistance,
- Order tracking,
- Feedback,
- Complaints, and
- Wallet management.

It reduces manual dependency and ensures faster resolution and higher customer satisfaction, making it operationally feasible.

3. Economic Feasibility

Since IBM Cloud and Watson Assistant provide free academic access, development cost was negligible.

The chatbot reduces the need for human agents, saving future operational costs.

It’s therefore economically viable both for educational and small-business applications.

4. Time Feasibility

The project was completed within the training period (June–July) under guided mentorship.

Watson’s modular structure enabled rapid development, testing, and deployment, proving it time-feasible for short-duration learning projects.

5. Educational Feasibility

The chatbot provided practical exposure to AI concepts, NLP, and cloud deployment, aligning perfectly with the IBM Generative AI course objectives.

It gave students a hands-on understanding of how conversational AI can be applied in real-world scenarios.

3.3 Tools / Technologies / Platform Used

Table. 3.3.1: Tools and Technologies Used.

Tool / Technology / Platform	Purpose and Description
Anaconda	Used to create a controlled environment for Python and AI libraries.

Tool / Technology / Platform	Purpose and Description
Jupyter Notebook	Platform for practicing AI and ML algorithms during initial training sessions.
Python	Programming language used for data analysis, ML model understanding, and experimentation.
Pandas / NumPy	Libraries for data manipulation, analysis, and mathematical computations.
Matplotlib / Seaborn	Libraries used for data visualization and plotting analytical results.
TensorFlow	Framework used to understand and simulate neural network concepts.
IBM Watson Assistant	Main tool for designing and building the chatbot using intents, entities, and dialog flows.
IBM Cloud	Deployment platform for hosting and testing the chatbot.
Watson Studio	Used for managing datasets, integrating Watson services, and monitoring performance.
Watson NLP Engine	Powers the chatbot’s natural language understanding for query interpretation.

3.4 Use Case Diagrams / Data Flow Diagrams

Case Diagram

Actors:

- User: Customer interacting with the chatbot.
- Admin: Maintains database, updates intents, and reviews feedback.

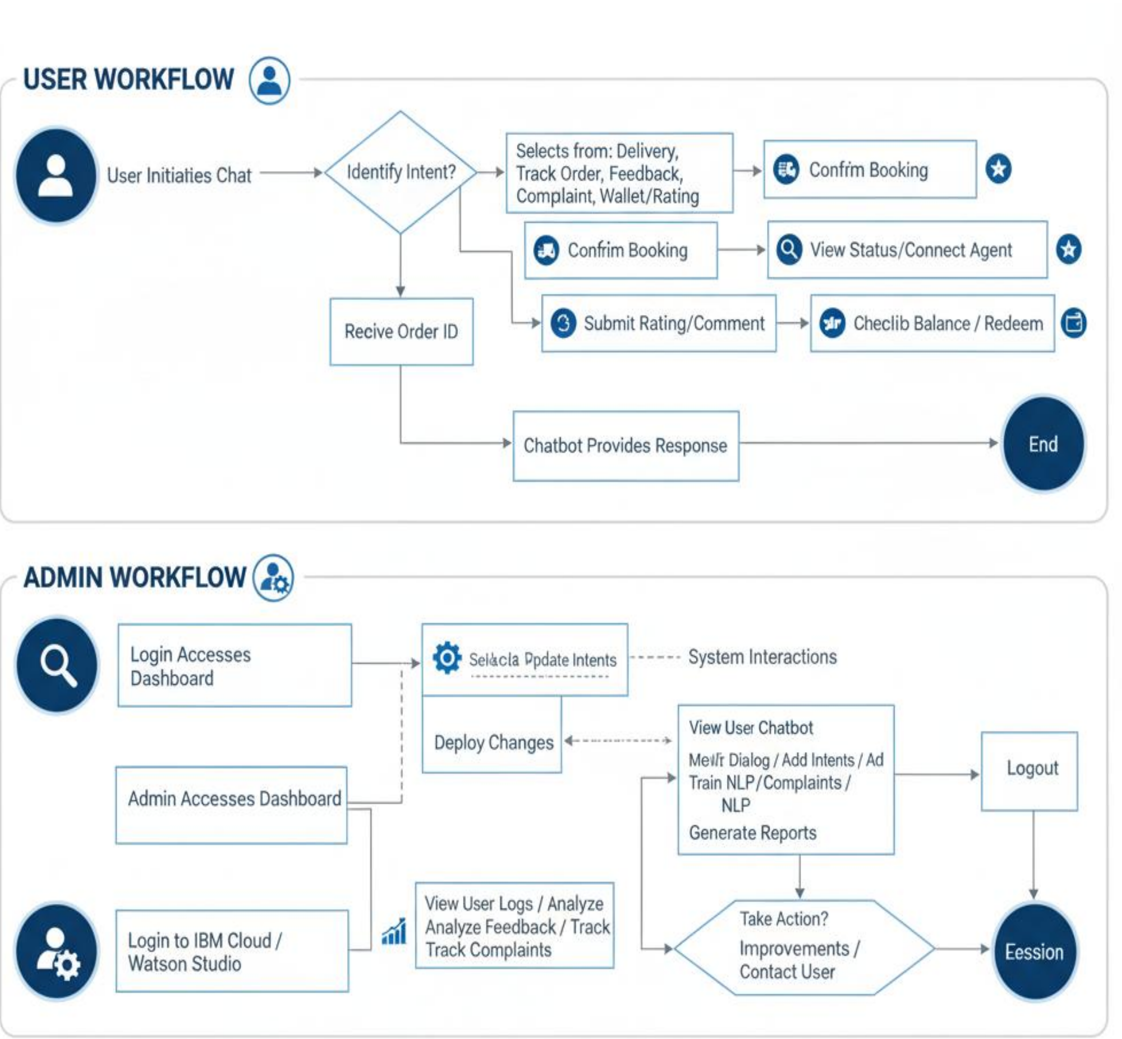

Figure 3.4.1: User and Admin Workflow of Porter's Chatbot.
Source: Image generated by Author using Google Gemini AI (2025)

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Cases:

- Delivery Assistance
- Track Order
- Feedback Collection
- Complaint Registration
- Wallet Points Inquiry
- Rating System

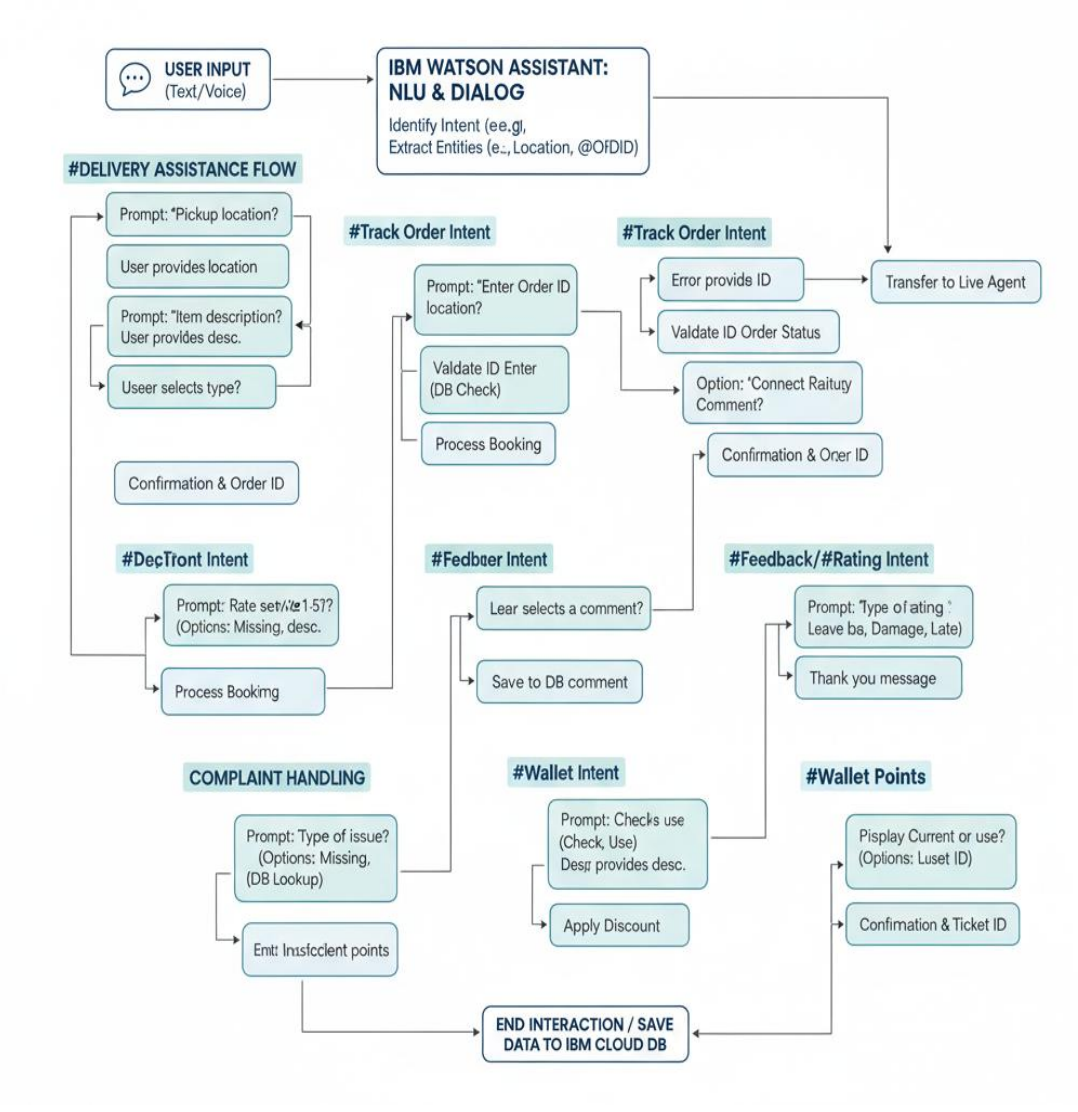

Figure 3.4.2: Case Flowchart.
Source: Image generated by Author using Google Gemini AI (2025).

Explanation:

1. User sends a query (e.g., *“Track my order”*).
2. Chatbot processes the input using Watson NLP.
3. Fetches or stores information in the IBM Cloud database.
4. Responds to the user instantly through a conversational interface.

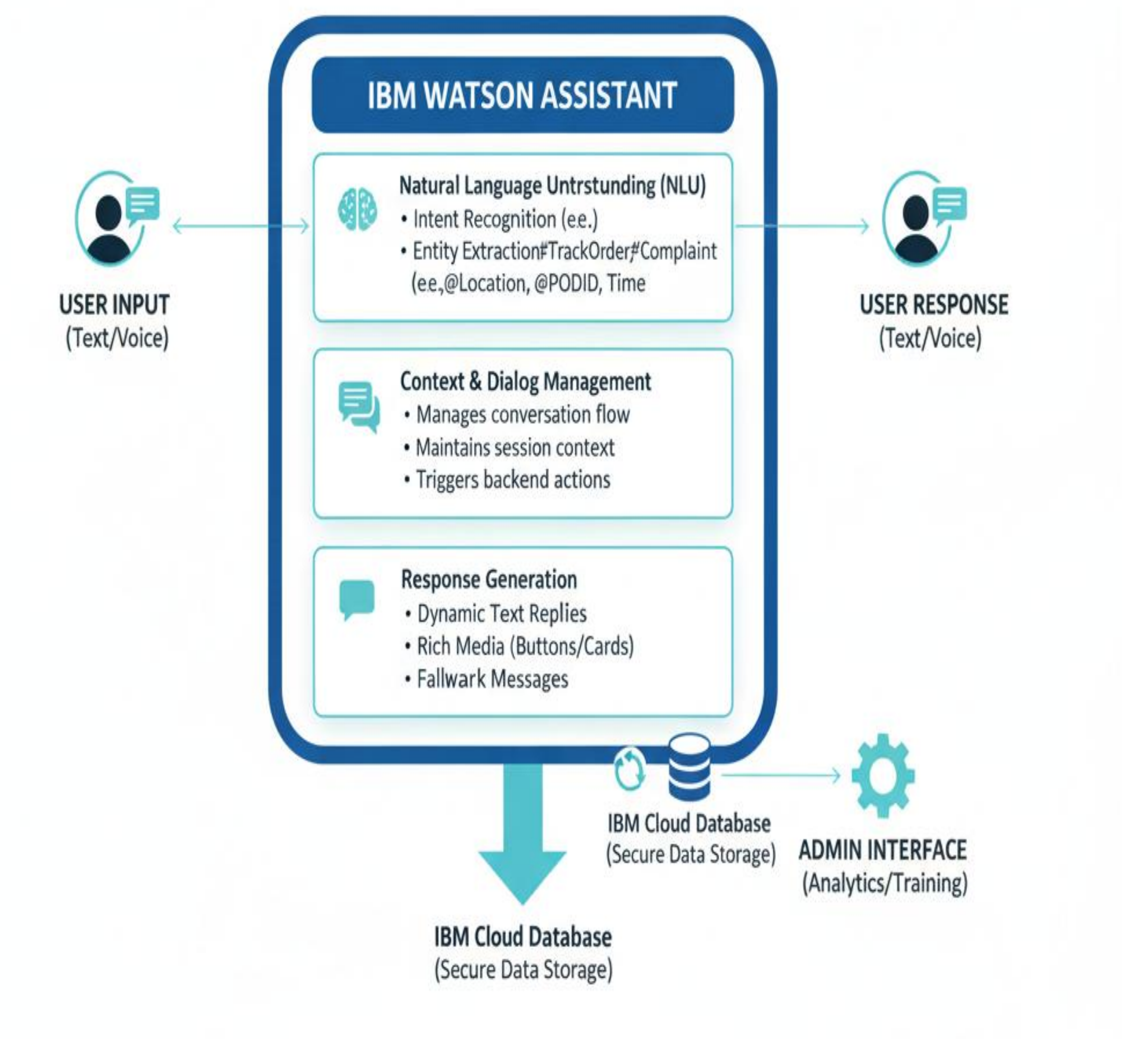

Figure 3.4.3: Watson AI Integration in Porter's Chatbot.
Source: Image generated by Author using Google Gemini AI (2025).

CHAPTER 4: PROJECT ACTIVITIES

4.1 Task Description

During the IBM Generative AI Summer Training (June–July), the team was tasked with conceptualizing and developing an AI-driven chatbot using IBM Watson Assistant. The objective was to explore how Artificial Intelligence can be applied to customer interaction automation, particularly in the logistics and delivery domain.

The project, titled "Porter Chatbot – The Porter Who Delivers", focused on creating a virtual assistant that helps customers book deliveries, track orders, handle complaints, provide feedback, and manage wallet points in a conversational way.

The work was divided into the following major phases:

1. Understanding Generative AI and Environment Setup
    - Installation and configuration of Anaconda and Jupyter Notebook.
    - Learning basic data handling using Pandas and NumPy.
    - Performing data visualization using Matplotlib and Seaborn.
    - Understanding the basics of Machine Learning algorithms such as Linear Regression, Logistic Regression, SVM, KNN, and Ensemble Learning.
2. Exploring AI Model Concepts
    - Introduction to TensorFlow and understanding neural networks (Feed-Forward Neural Networks).
    - Learning clustering techniques such as K-Means and Hierarchical Clustering.
3. Introduction to IBM Watson and Cloud Services
    - Logging into IBM Cloud and accessing Watson Assistant.
    - Learning how Watson uses Natural Language Processing (NLP) to identify intents and generate contextual responses.
4. Project Design and Implementation (Porter Chatbot)
    - Designing the chatbot flow for five key functionalities:
        1. Delivery Support
        2. Order Tracking
        3. Complaint Handling
        4. Feedback Collection

5. Wallet Points and Rating System
    - Creating intents, entities, and dialog nodes in Watson Assistant.
    - Testing the chatbot for various user queries and refining responses.
    - Hosting and monitoring performance through IBM Cloud.

5. Testing and Evaluation
    - Running multiple test scenarios to check the chatbot's accuracy.
    - Measuring response speed, intent detection, and error handling.
    - Making adjustments to ensure smooth and natural interaction flow.
6. Documentation and Presentation
    - Preparing the complete project documentation.
    - Presenting the project outcomes and learning during the final evaluation session.

4.2 Tools and Technologies Used for Porter Chatbot

Table 4.2.1: Tools and Technologies.

Tool / Technology / Platform	Purpose and Description
Anaconda	Used to create a controlled environment for Python and AI libraries.
Jupyter Notebook	Platform for practicing AI and ML algorithms during initial training sessions.
Python	Programming language used for data analysis, ML model understanding, and experimentation.
Pandas / NumPy	Libraries for data manipulation, analysis, and mathematical computations.
Matplotlib / Seaborn	Libraries used for data visualization and plotting analytical results.

Tool / Technology / Platform	Purpose and Description
TensorFlow	Framework used to understand and simulate neural network concepts.
IBM Watson Assistant	Main tool for designing and building the chatbot using intents, entities, and dialog flows.
IBM Cloud	Deployment platform for hosting and testing the chatbot.
Watson Studio	Used for managing datasets, integrating Watson services, and monitoring performance.
Watson NLP Engine	Powers the chatbot's natural language understanding for query interpretation.

4.3 Technical Application

The Porter Chatbot project served as a hands-on application of various AI and machine learning techniques learned during the training.

1. Data Analysis and Manipulation:
   Using Pandas and NumPy, the team practiced how real-world delivery datasets can be structured, cleaned, and analyzed.
2. Machine Learning Understanding:
   - Explored algorithms like Linear Regression, Logistic Regression, and KNN, which helped in understanding how predictive models are built.
   - Applied Ensemble Learning concepts such as Bagging and Boosting to comprehend model optimization.
3. Clustering and Neural Networks:
   - Understood K-Means Clustering for grouping similar data points.
   - Explored Feed-Forward Neural Networks using TensorFlow to understand how AI models learn patterns.
4. AI Integration through Watson Assistant:

- Used Watson Assistant to create conversational flows and train the chatbot to understand delivery-related intents.
- Defined entities like delivery location, tracking ID, feedback message, and complaint type.
- Configured dialog nodes to connect intents and ensure logical conversation paths.

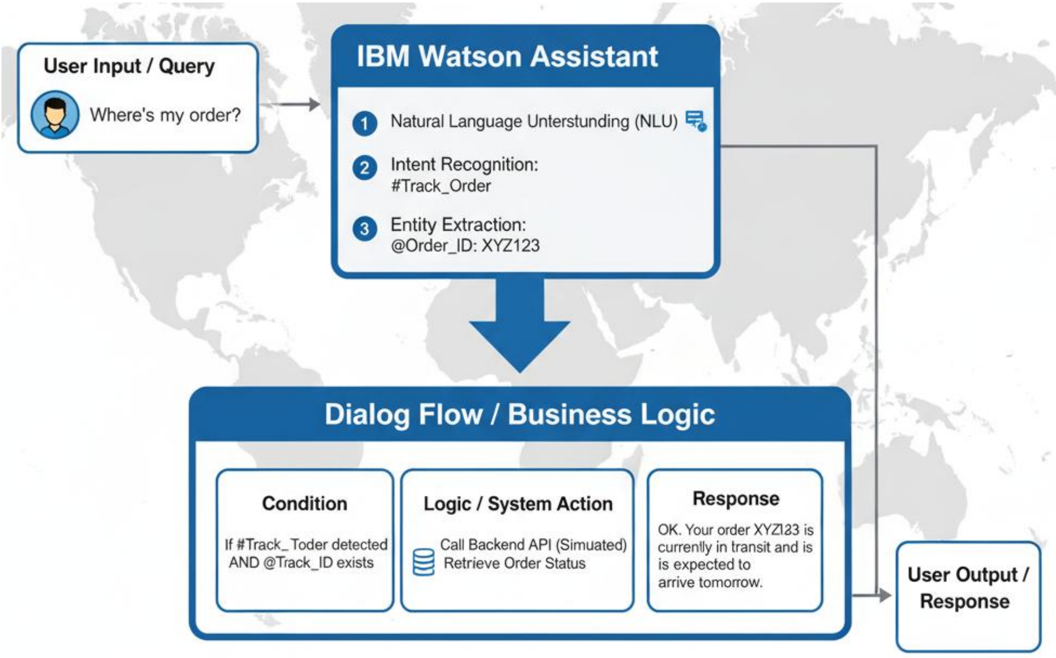

Figure 4.3.1: Watson Assistant Dialog Flow Diagram.
Source: Image generated by Author using Google Gemini AI (2025).

5. Cloud Deployment and Accessibility:
   - The final chatbot was hosted on IBM Cloud, making it accessible via browser for testing.
   - This provided practical exposure to cloud-based AI deployment.
6. Testing and Validation:
   - Tested different user inputs to ensure intent recognition accuracy.
   - Refined Watson responses to improve naturalness and relevance.

Through this process, students successfully applied their theoretical AI learning to a working real-world solution.

4.4 Challenges Faced

Every project comes with its set of challenges. During the development of The Porter Chatbot, the team encountered several technical and conceptual hurdles:

1. Understanding AI Concepts:
   Initially, grasping advanced AI topics such as neural networks and ensemble methods was challenging. The team had to revise foundational mathematics and ML concepts.
2. Setting Up IBM Cloud:
   Logging into IBM Cloud and integrating services with Watson Assistant required understanding cloud environment structures and permissions.
3. Designing Conversational Flows:
   Creating logical, user-friendly dialog nodes without loops or broken connections demanded attention to detail and multiple revisions.
4. Intent Training Accuracy:
   Some user inputs were not recognized correctly at first. The team expanded the training phrases to improve the accuracy of Watson's NLP.
5. Testing and Debugging:
   Continuous testing was required to ensure that every intent, from complaint registration to feedback collection, triggered correctly.
6. Time Constraints:
   Balancing theoretical learning (AI algorithms) and practical implementation (chatbot building) within the limited training period was a key challenge.

Table 4.4.1: Project Challenges and Solutions.

Challenge Faced	Action / Solution Implemented	Outcome
Understanding AI Concepts	Revised foundational mathematics and core ML concepts.	Improved grasp of advanced topics like neural networks and ensemble methods.

Challenge Faced	Action / Solution Implemented	Outcome
Setting Up IBM Cloud	Learned cloud environment structures, permissions, and service integration.	Successfully integrated IBM Cloud services with Watson Assistant.
Designing Conversational Flows	Conducted multiple revisions and focused on logical connection between dialog nodes.	Created logical, user-friendly dialog nodes without loops or broken connections.
Intent Training Accuracy	Expanded the training phrases (utterances) in Watson Assistant.	Improved the accuracy and reliability of Watson's Natural Language Processing (NLP).
Time Constraints	Focused effort and effective time management was applied to balance learning and implementation.	Successfully completed both theoretical learning and practical chatbot building within the limited period.

Despite these hurdles, teamwork, instructor guidance, and consistent effort helped successfully complete the project.

CHAPTER 5: LEARNING AND DEVELOPMENT

The summer training on Generative Artificial Intelligence, conducted by IBM from June to July, was a transformative experience that bridged the gap between theoretical learning and real-world technical applications. The training provided me with deep insights into data science, machine learning, artificial intelligence, and modern cloud-based technologies. Working under professional mentors and instructors allowed me to not only gain technical competence but also develop essential soft skills such as teamwork, problem-solving, and communication. The culmination of this learning experience was the development of an AI-powered chatbot project titled “Porter’s ChatBot (The Porter Who Delivers)”, which demonstrated the integration of AI into a practical service-based application.

5.1 Skills and Knowledge Gained During the Training

The training began with a strong foundation in setting up the technical environment using Anaconda and Jupyter Notebook, which served as the main development platforms throughout the course. I learned how to install, configure, and efficiently use these tools for executing Python-based data science projects. Through guided sessions, I became familiar with the Python programming language, especially in the context of data analysis, data visualization, and machine learning.

The first phase of learning focused on understanding Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. These libraries allowed me to manipulate large datasets, perform data cleaning and transformation, and visualize data trends effectively. I learned to perform data preprocessing, which is the most critical step in building reliable AI models. Through various exercises, I explored how real-world datasets are structured, analyzed, and interpreted for predictive modeling.

Next, we studied several machine learning algorithms, including Linear Regression, Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Ensemble Learning Techniques such as Bagging, Boosting, and Random Forest. Implementing these algorithms in Jupyter Notebook gave me a practical understanding of supervised learning and model performance evaluation using metrics like accuracy, precision, and recall. Later, we explored K-Means Clustering and Hierarchical Clustering, which introduced the concept of unsupervised learning. This knowledge helped me understand how clustering can identify meaningful patterns without labeled data.

A crucial part of the training was an introduction to TensorFlow, a powerful framework for deep learning and neural networks. I learned about the architecture of neural networks, including feed-forward propagation, activation functions, and optimization techniques. This enhanced my conceptual clarity about how intelligent systems learn and adapt over time.

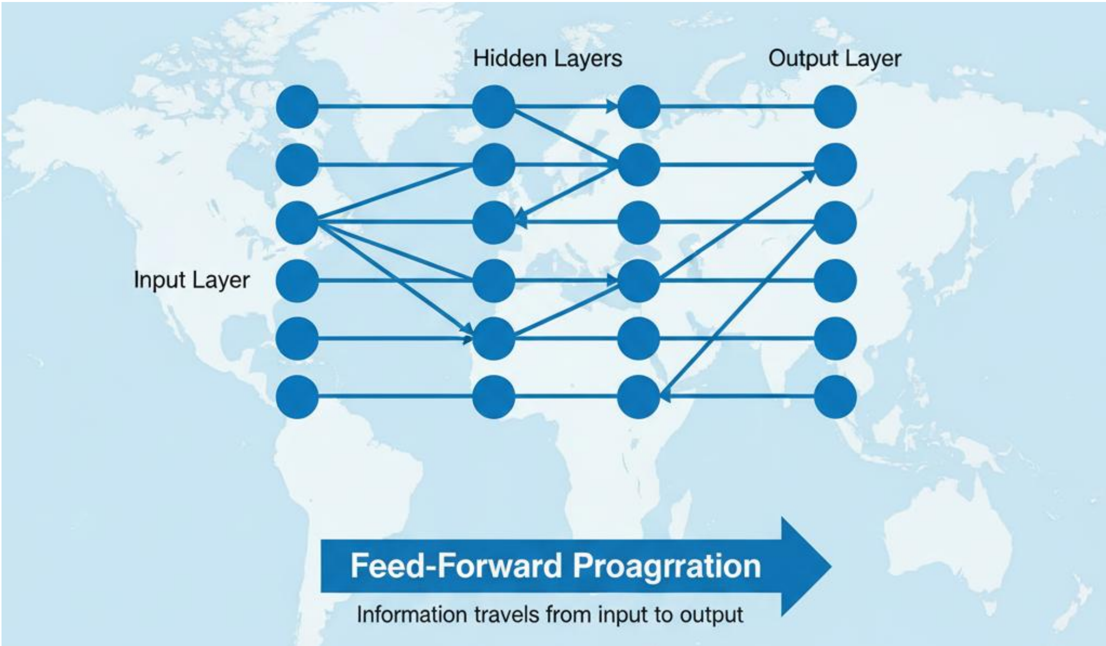

Figure 5.1.1: Conceptual Model of Neural Network (Feed-Forward propagation).
Source: Image generated by Author using Google Gemini AI (2025)

In the later sessions, we were introduced to IBM Watson AI and the IBM Cloud platform, where we learned to design and deploy AI models and chatbots. We were guided through the process of logging into the IBM Cloud environment, setting up Watson Assistant, and creating conversational flows. This practical exposure helped me understand how modern AI systems work in real-time applications.

Table 5.1.1: Technical Skills Gained.

Category	Skills & Tools Learned	Example Application
Programming & Data Libraries	Python, Anaconda, Jupyter Notebook, **Pandas, NumPy, Matplotlib, Seaborn**	Data manipulation, visualization, and environment setup.
Machine Learning (ML)	Linear/Logistic Regression, SVM, KNN, K-Means Clustering, Hierarchical Clustering	Supervised and Unsupervised model implementation and evaluation.
Advanced ML & Deep Learning	**Ensemble Techniques** (Bagging, Boosting, Random Forest), **TensorFlow**, Neural Networks	Enhancing model accuracy; understanding feed-forward propagation.
AI/Cloud Platform	**IBM Watson Assistant**, **IBM Cloud**	Designing, training, and deploying a functional AI chatbot.

The final part of the training involved applying all these learnings to a practical project, *Porter's ChatBot (The Porter Who Delivers)* — a chatbot designed to assist users in managing delivery requests, handling complaints, collecting feedback, rating services, and tracking orders. By designing and training this chatbot, I gained valuable experience in intent recognition, entity extraction, dialogue management, and AI-based automation workflows. Overall, the training developed my proficiency in data analytics, machine learning model design, and AI-driven application development, while improving my logical and analytical skills.

5.2 Teamwork, Communication, and Problem-Solving Skills

In addition to technical expertise, the training program and project work significantly improved my interpersonal and professional skills. During the development of *Porter's*

ChatBot, I worked closely with other trainees, collaborating on problem-solving, brainstorming ideas, and managing different aspects of the project. This collaboration taught me the importance of effective teamwork and coordination in a real project environment.

Table 5.2.1: Professional Skills Development.

Skill Area	**Demonstrated Application in Project Work**	**Outcome**
Teamwork & Collaboration	Working with trainees to divide tasks, brainstorm, and integrate components of the **Porter ChatBot**.	Improved coordination and effective system integration.
Problem-Solving	**Systematically handling errors**, data inconsistencies, and compatibility issues encountered during implementation.	Enhanced critical thinking and troubleshooting abilities.
Communication	Explaining complex technical concepts and project progress during discussions and review sessions.	Improved clarity in both verbal and written technical explanations.
Adaptability & Time Management	Working under project deadlines while simultaneously learning and applying new technologies.	Increased organization, focus, and proactive work ethic.

I learned how to divide tasks based on team members' strengths, set deadlines, and integrate each component into a unified system. Through discussions and review sessions, I developed communication skills both verbal and written that were essential for explaining complex technical concepts in a simple manner to instructors and peers.

Problem-solving was another area where I grew immensely. While implementing machine learning algorithms or integrating Watson Assistant, I encountered various errors, data inconsistencies, and compatibility issues. By researching solutions, testing multiple approaches, and seeking guidance from mentors, I learned how to handle technical challenges systematically and with patience. This process enhanced my critical thinking and troubleshooting abilities, which are key qualities for any aspiring engineer.

Furthermore, the experience taught me the value of adaptability and time management. Working under deadlines while learning new technologies encouraged me to stay organized, focused, and proactive. These skills will continue to help me in future academic projects and professional environments.

5.3 Certifications Received During the Internship Period

At the completion of the training, I received a Certificate of Completion from IBM, recognizing my successful participation in the Summer Training on Generative Artificial Intelligence. This certification serves as proof of my exposure to real-world AI tools and technologies and my ability to apply them in practical contexts. The training covered a broad spectrum of learning outcomes, including:

- Proficiency in Python for Data Science and AI
- Understanding of Data Visualization and Data Manipulation Techniques
- Application of Machine Learning and Deep Learning Models
- Exposure to TensorFlow Framework for neural networks
- Hands-on experience with IBM Watson Assistant and IBM Cloud for AI chatbot development

Additionally, the training sessions included practical assignments, quizzes, and guided projects that simulated industrial problem-solving scenarios. Completing these tasks helped me build a strong technical foundation and confidence in applying AI tools independently.

The certificate from IBM not only adds value to my academic portfolio but also reflects my ability to engage with advanced technologies and adapt to the evolving landscape of Artificial Intelligence. This training has strengthened my readiness to take on future challenges in the AI and data science domains.

Figure 5.3.1: IBM Certificate.

CHAPTER 6: SUMMARY AND CONCLUSION

The Porter Chatbot project was a complete learning experience integrating Artificial Intelligence, Machine Learning, and Cloud Computing concepts into one functional application.

Through this project, students learned how AI-based systems can replace or assist traditional customer service channels. The chatbot successfully handled five major service areas — Delivery Support, Order Tracking, Feedback Collection, Complaint Handling, and Wallet Points Management.

The training helped bridge the gap between theory and practice. Concepts such as data manipulation, regression models, ensemble learning, and clustering were applied to real-world AI applications like Watson Assistant.

Table 6.1: Summary of Chatbot Services Areas.

Theoretical Concept Covered in Training	Practical Application in Porter Chatbot Project
Artificial Intelligence (AI) / NLP	Simulation of intelligent human conversation; accurate Intent and Entity recognition.
Machine Learning (ML) / Data Manipulation	Data cleaning and labeling of training phrases for NLU model accuracy.
Cloud Computing	Deployment and hosting of the conversational AI solution on **IBM Cloud**.
Regression/Classification Models	Implicit foundation for the core **NLU model** within Watson Assistant.
End-to-End Workflow	Navigating the entire AI lifecycle: design, training, deployment, and testing.

The IBM Summer Training not only improved technical proficiency but also inspired students to pursue careers in AI and Data Science.

In conclusion, the project achieved its goal — to understand, design, and deploy a functional AI-based chatbot capable of simulating intelligent human conversation and assisting users in delivery-related tasks. It stands as a testimony to the growing role of Generative AI in business automation and smart communication systems.

The Porter Chatbot project was a transformative, end-to-end learning experience that successfully integrated several cutting-edge fields of computer science: Artificial Intelligence (AI), Machine Learning (ML), and Cloud Computing. Far from being a theoretical exercise, this project culminated in a functional application designed to solve a tangible real-world business challenge: optimizing customer support for a delivery service.

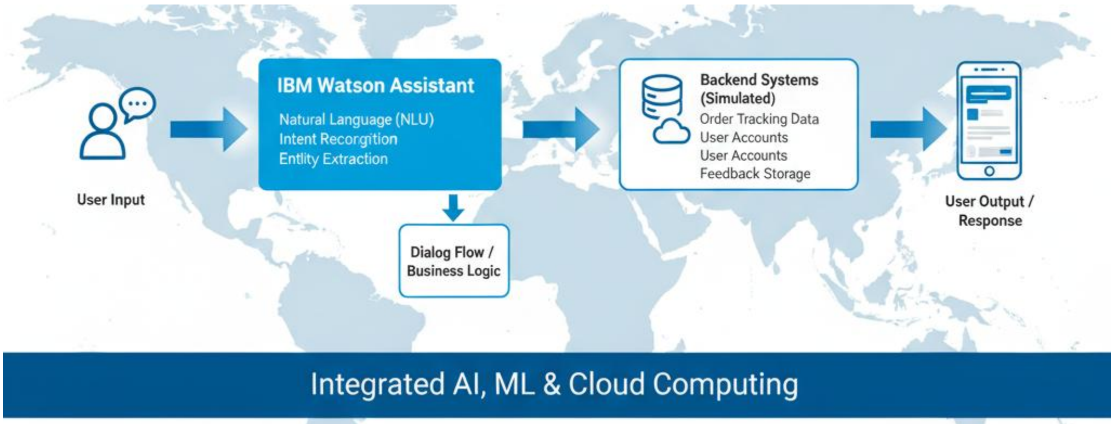

Figure 6.1: End-to-End Porter Chatbot Architecture.
Source: Image generated by Author using Google Gemini AI (2025).

By leveraging IBM Watson Assistant, the project team navigated the entire lifecycle of a conversational AI solution, from initial design and data labeling to deployment and functional testing. The project demonstrated how intelligent, AI-based systems can be deployed not only to replace but also to significantly assist traditional customer service channels, thereby enhancing efficiency and customer experience.

The primary objective of the Porter Chatbot project was to understand, design, and deploy a functional AI-based chatbot capable of simulating intelligent human conversation and automating essential delivery-related tasks. This goal was conclusively achieved. The chatbot was successfully trained and configured to manage and respond to user queries across five major service areas, demonstrating its versatility and robust conversational architecture:

1. Delivery Support: Handling new delivery requests, service inquiries, and product information.
2. Order Tracking: Providing real-time status updates using simulated backend data.
3. Feedback Collection: Prompting users for input on their experience and efficiently classifying sentiment.
4. Complaint Handling: Triage and automatic routing of serious issues while resolving minor queries instantly.
5. Wallet Points Management: Providing account balance information and explaining rewards programs.

Table 6.2: Key Learning Outcomes and Application.

Service Area	**Primary Function in Porter Chatbot**	**Versatility Demonstrated**
Delivery Support	Handling new delivery requests, service inquiries, and product information.	Versatility in general customer inquiry management.
Order Tracking	Providing real-time status updates using simulated backend data.	Integration capability with external data sources (APIs).
Feedback Collection	Prompting users for input on their experience and efficiently classifying sentiment.	Data collection and preliminary NLU/Sentiment analysis.
Complaint Handling	Triage and automatic routing of serious issues while resolving minor queries instantly.	Automation of critical customer service workflow.

Service Area	Primary Function in Porter Chatbot	Versatility Demonstrated
Wallet Points Management	Providing account balance information and explaining rewards programs.	Integration with user account/financial data (simulated).

The ability of the chatbot to accurately recognize Intents (the user's goal, e.g., #Track_Order) and extract Entities (the necessary data, e.g., @Order_ID) from natural language input validates the effectiveness of the training data and the chosen NLU framework. The successful implementation of a coherent dialog flow confirms the project's technical competence in building structured, goal-oriented conversations.

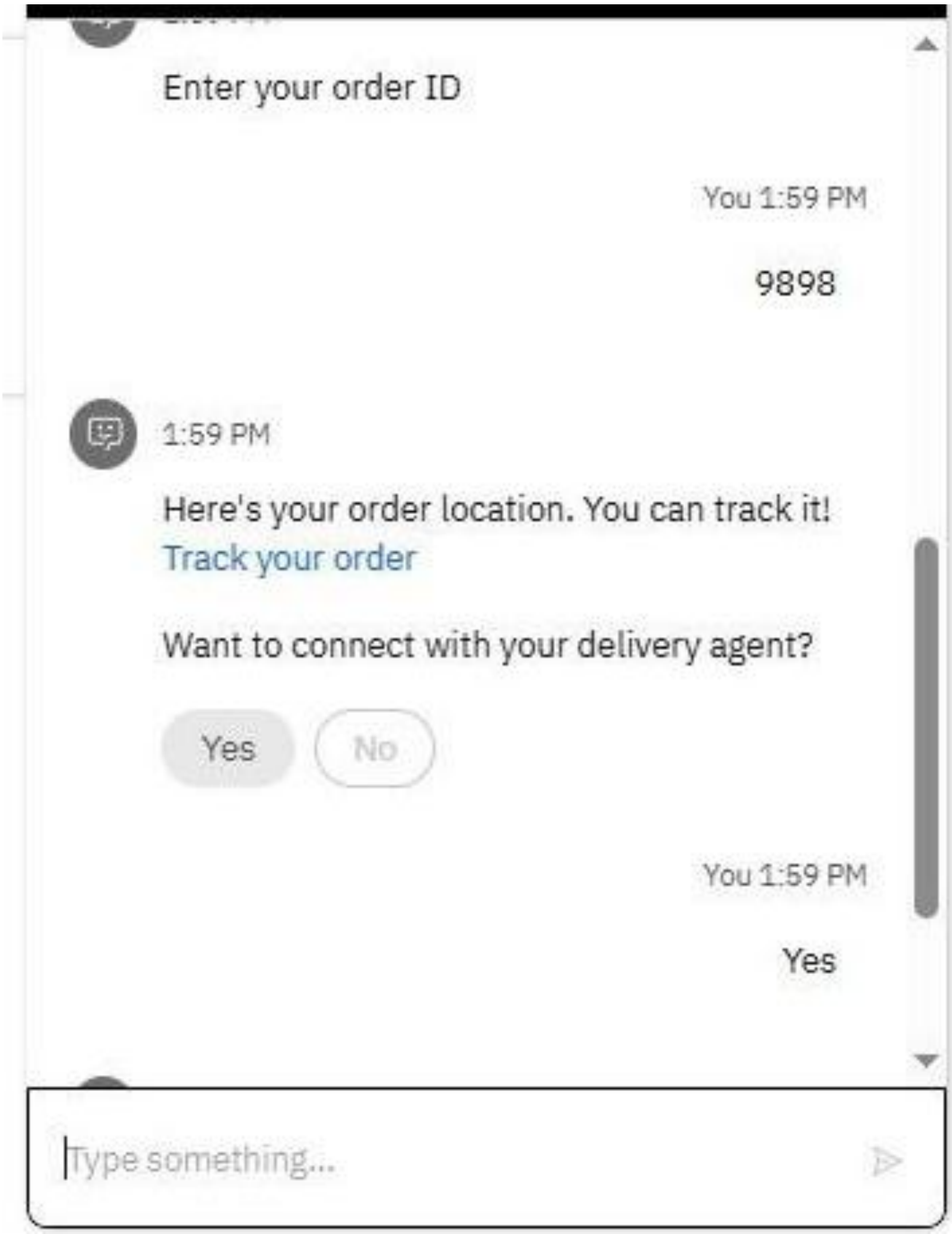

Figure 6.2: Oder Tracking Intent Recognition.
Source: Image generated by Author using Google Gemini AI (2025).

CHAPTER 7: SUGGESTIONS FOR IMPROVEMENT

One of the major areas of improvement could be the inclusion of more hands-on practical sessions. Although the training provided an excellent theoretical foundation on concepts like data visualization, regression techniques, support vector machines, and clustering, students could benefit from additional live practice sessions on Jupyter Notebook or IBM Cloud. Practical exercises conducted alongside lectures would help learners apply the concepts immediately, identify their errors, and strengthen their problem-solving skills. Moreover, it would be beneficial if participants were given access to more real-world and large-scale datasets instead of small, simplified examples. Working with such datasets would help in understanding the challenges of real data, including missing values, outliers, and preprocessing issues, which are essential skills for any aspiring data scientist or AI engineer.

Table 7.1: Proposed Enhancements to Training Methodologies.

Area of Improvement	Suggested Enhancement	Rationale for Inclusion
Hands-on Practice	**Additional Live Practical Sessions** (using Jupyter Notebook/IBM Cloud).	Allows learners to immediately apply concepts, identify errors, and strengthen **problem-solving skills** alongside lectures.
Dataset Relevance	Provide access to **more real-world and large-scale datasets**.	Helps students understand the challenges of real data, including **missing values, outliers, and preprocessing issues**, which are essential industry skills.

Another improvement could be extending the duration of the training. The course was held from June to July, which provided only a limited window to cover a vast syllabus that included machine learning algorithms, data manipulation techniques, ensemble learning, clustering methods, and introduction to deep learning and neural networks.

An additional week or two would allow participants to explore each topic in more depth and gain a more confident grasp of implementation. It would also allow more time for doubt-clearing sessions, mini-projects, and interactive discussions. Additionally, more structured project guidance would make the learning outcome stronger. While our final project—developing a chatbot for delivery support, feedback collection, and customer complaint handling—was a great opportunity to integrate various AI tools, more step-by-step mentorship during project development and evaluation could ensure that every participant fully understands both the conceptual and technical aspects of their solution. A formal evaluation system, including grading criteria and detailed feedback from mentors, would further motivate students and highlight areas where they could improve.

The inclusion of small-group mentoring sessions would also significantly improve the overall learning experience. Sometimes, due to large batch sizes, it becomes difficult for every student to get personalized attention or have their specific doubts resolved during live classes. Dividing participants into smaller groups, each assigned a mentor or teaching assistant, would promote more direct communication, allow focused discussions, and encourage collaborative learning among peers. Another key area for enhancement could be the introduction of short modules on AI ethics, data privacy, and model bias, which are increasingly becoming essential components of responsible AI development. Understanding these ethical implications ensures that students not only build technically sound models but also develop a sense of social and professional responsibility as future technologists.

Furthermore, providing extended access to the IBM Cloud platform and Watson AI tools even after the completion of the training would be extremely helpful. Many students, after completing the course, wish to explore and practice further but lose access to the platform once the training period ends. Granting an extended or limited post-training license would allow participants to continue their learning journey independently and explore advanced use cases of IBM's AI technologies. It would also be beneficial to introduce an online discussion forum or community platform where learners can share knowledge, ask queries, and interact with mentors beyond class hours. Such a support system would foster continuous engagement and peer-to-peer learning. The course could also include a short orientation session before the actual training begins, where participants are introduced to the basic Python environment, libraries like Pandas and NumPy, and the interface of Jupyter Notebook. This would ensure that everyone starts at the same level, especially for students who are completely new to programming or data analysis.

Figure 7.1: Conceptual Diagram of AI Ethics Integration.
Source: Image generated by Author using Google Gemini AI (2025).

From a participant's perspective, students themselves can also contribute to improving the learning outcome by preparing in advance and staying actively engaged throughout the training. Future participants should be encouraged to revise the fundamentals of Python, statistics, and data structures before the course begins so that they can grasp machine learning algorithms more efficiently. Regular note-taking, self-practice, and experimenting with different datasets during the course can also make a big difference in understanding. Students should actively participate during live sessions, ask questions, and collaborate with peers on small coding exercises. These habits not only enhance technical knowledge but also improve communication, teamwork, and critical thinking skills, which are essential in any real-world AI project.

Table 7.2: Future Participant Preparation Guide.

Participant Responsibility Category	Specific Actions for Improvement	Expected Learning Outcome & Benefits
Pre-Course Preparation	**Revise fundamentals** of Python, statistics, and	Enables participants to grasp machine learning

Participant Responsibility Category	Specific Actions for Improvement	Expected Learning Outcome & Benefits
	data structures before the course begins.	algorithms more efficiently and start at a common level.
Active Engagement During Course	**Regular note-taking, self-practice,** and experimenting with different datasets during the course.	Makes a significant difference in understanding, reinforces technical knowledge, and develops practical skills.
Collaboration & Participation	**Actively participate** during live sessions, ask questions, and collaborate with peers on coding exercises.	Enhances communication, teamwork, and critical thinking skills, crucial for real-world AI projects.

Overall, the training was exceptionally well-structured and delivered, with experienced instructors, engaging sessions, and a balance of theory and application. However, implementing the above suggestions could further elevate the learning quality of the program. By making the sessions more interactive, extending practice time, focusing on real-world datasets, and introducing mentorship and evaluation systems, IBM can ensure that participants not only learn the tools and techniques but also gain a deeper understanding of how these technologies function in industry contexts. Additionally, incorporating post-training access and discussion platforms would promote continuous learning even after completion of the course. Such refinements would make the Generative AI Summer Training more comprehensive, learner-centered, and impactful. These enhancements would not only benefit future students but would also strengthen IBM's reputation for delivering high-quality, industry-relevant technical training programs that effectively bridge the gap between academic learning and professional application.

While the project was successful, certain improvements could further enhance its quality and performance:

1. Add Voice Integration:
   Integrate speech-to-text and text-to-speech modules for hands-free operation and accessibility.
2. Multilingual Support:
   Include regional languages to cater to a wider audience across different areas of India.
3. Advanced Tracking Integration:
   Connect the chatbot with real-time GPS APIs to provide accurate parcel location data.
4. Payment Gateway Integration:
   Enable customers to pay delivery charges directly through the chatbot interface.
5. User Dashboard:
   Develop a small dashboard to visualize wallet points, feedback history, and order statistics.
6. Machine Learning-based Feedback Analysis:
   Use sentiment analysis on collected feedback to automatically rate customer satisfaction levels.
7. Continuous Model Training:
   Regularly update intents and expand the dataset to improve chatbot understanding and performance accuracy.

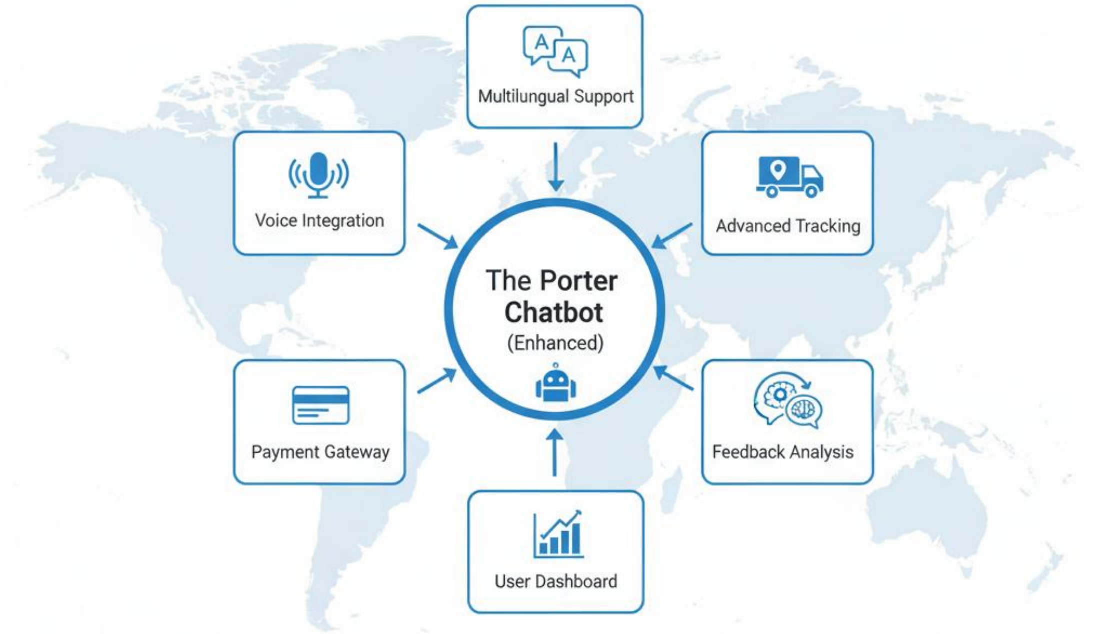

Figure 7.2: Enhanced Porter's Chatbot Roadmap.
Source: Image generated by Author using Google Gemini AI (2025).

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APPENDIX

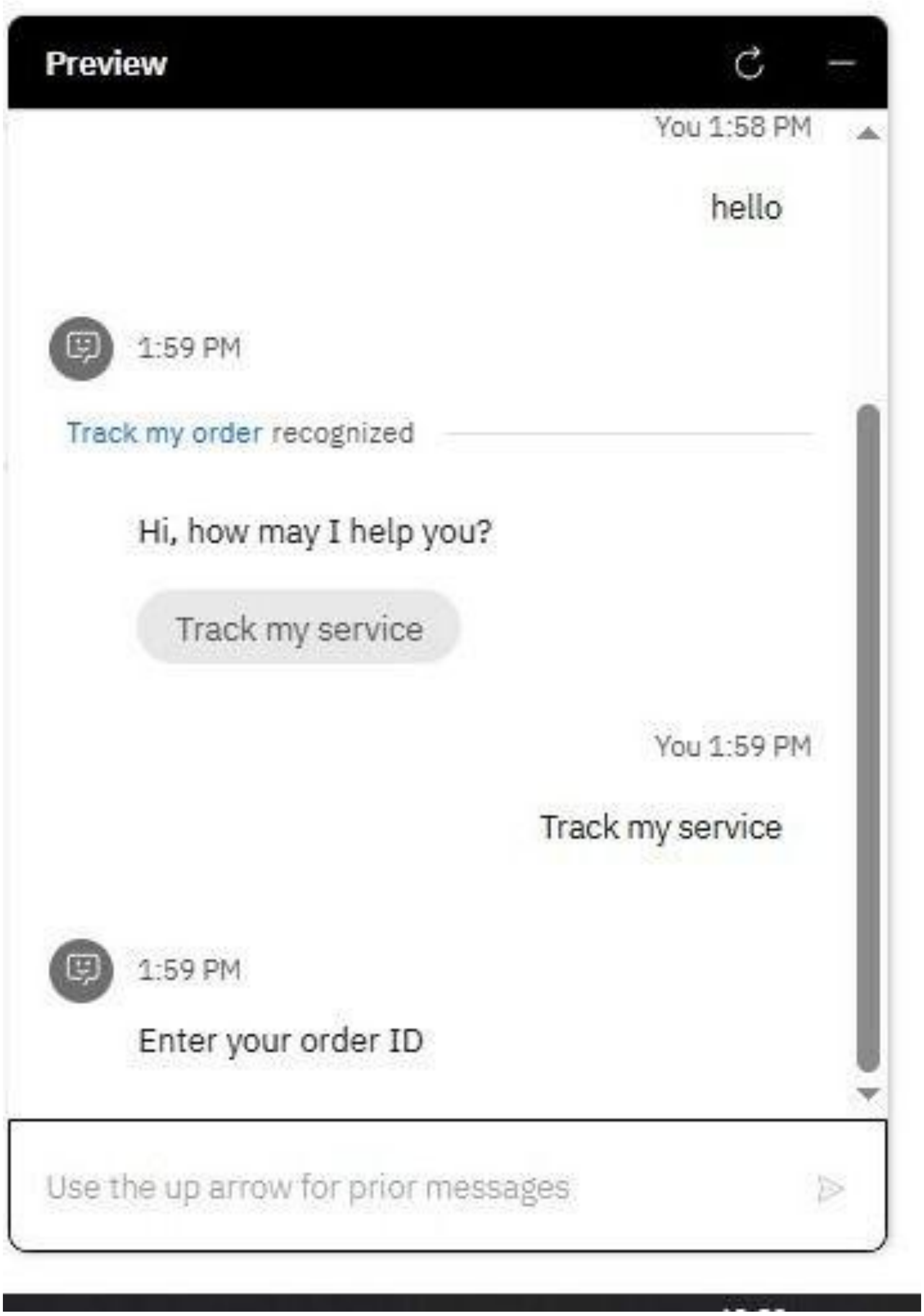
Appendix A – Sample Chat Flow of Porter Chatbot

Example conversation:

User: Hello!

Bot: Hi, How may I help you?

User: Track my service.



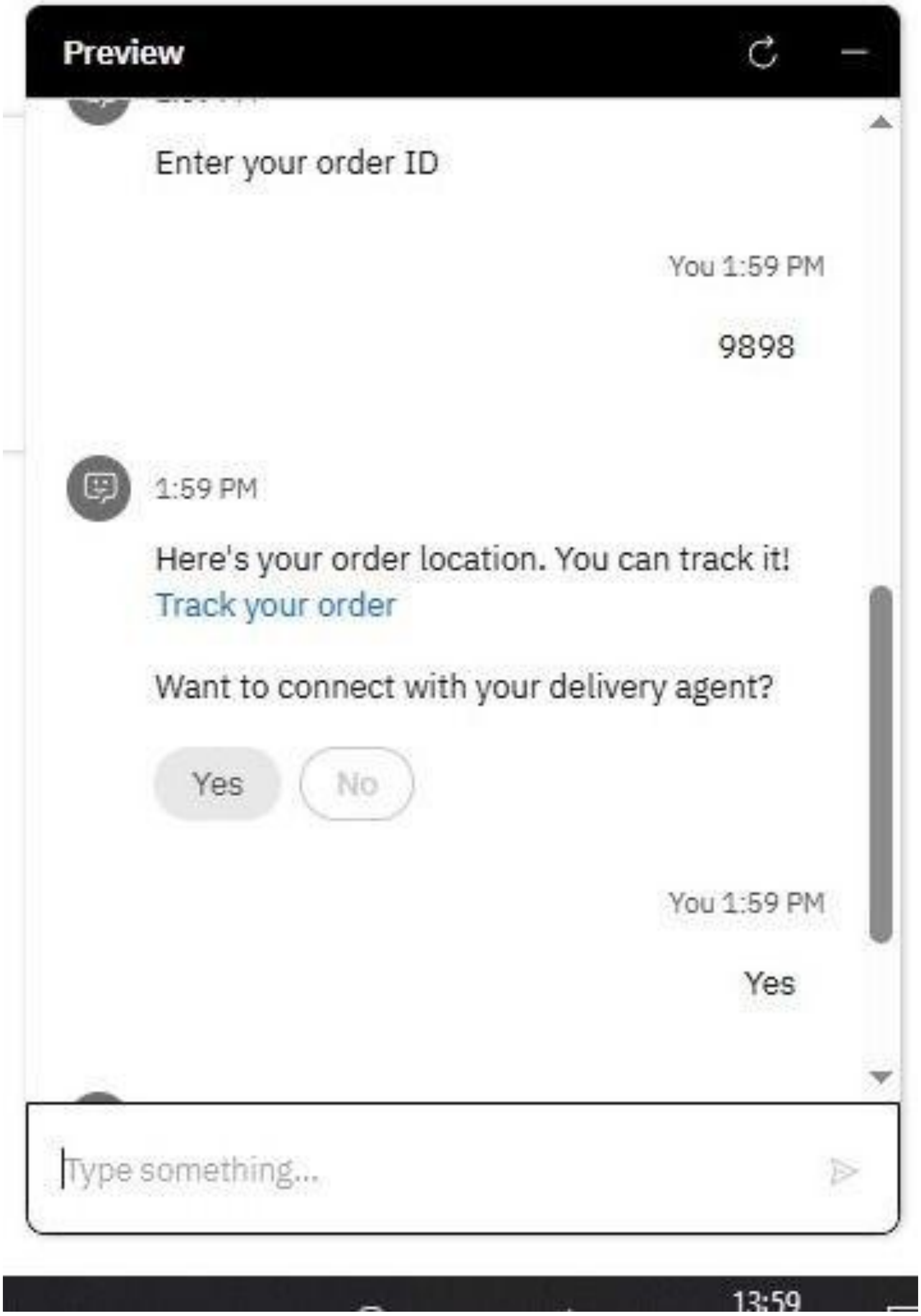
Bot: Please enter your tracking ID.

User: 45872DEL

Bot: Here's your oder location. You can track it! *Track your oder*

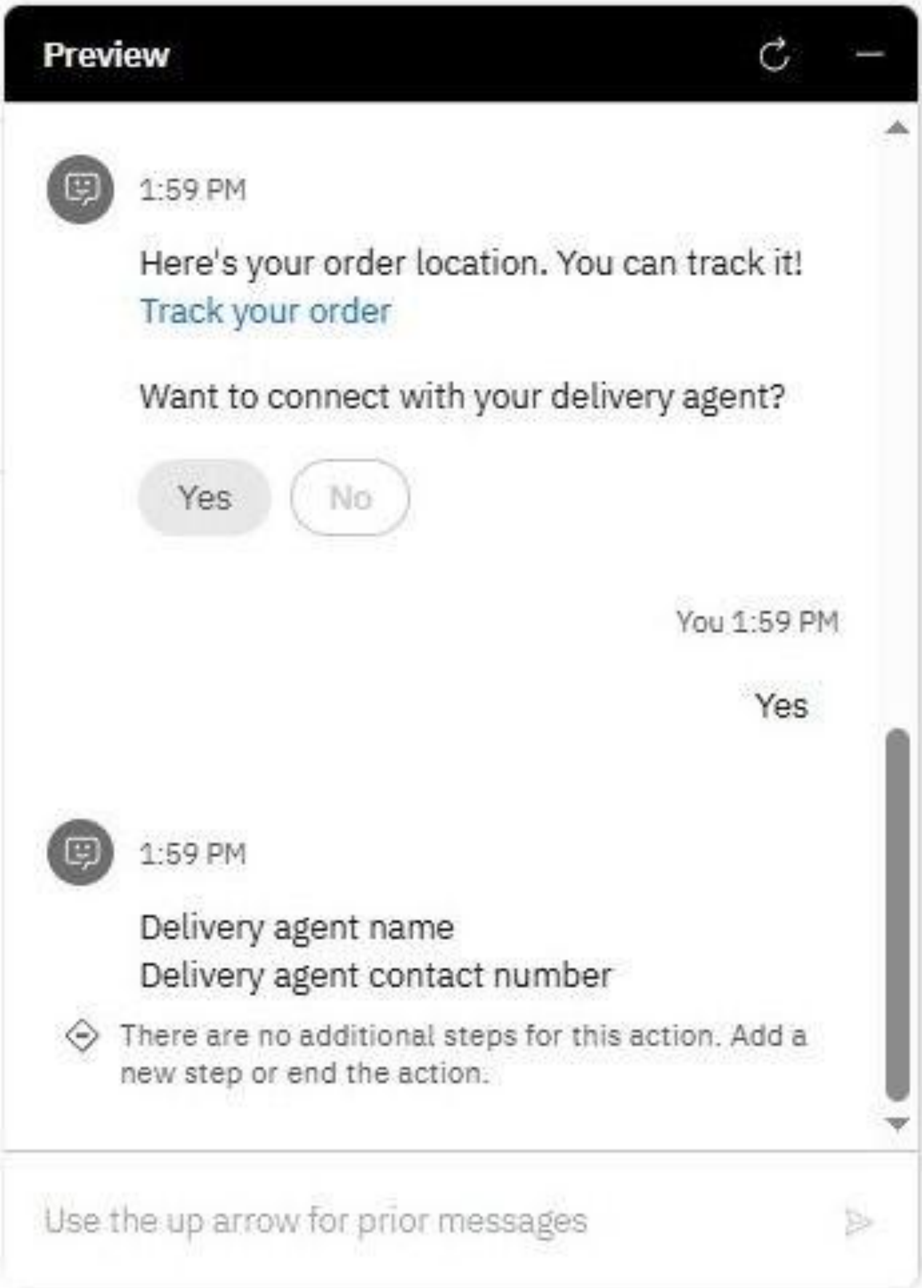
Want to connect with your delivery agent?

User: Yes



Bot: Delivery agent name

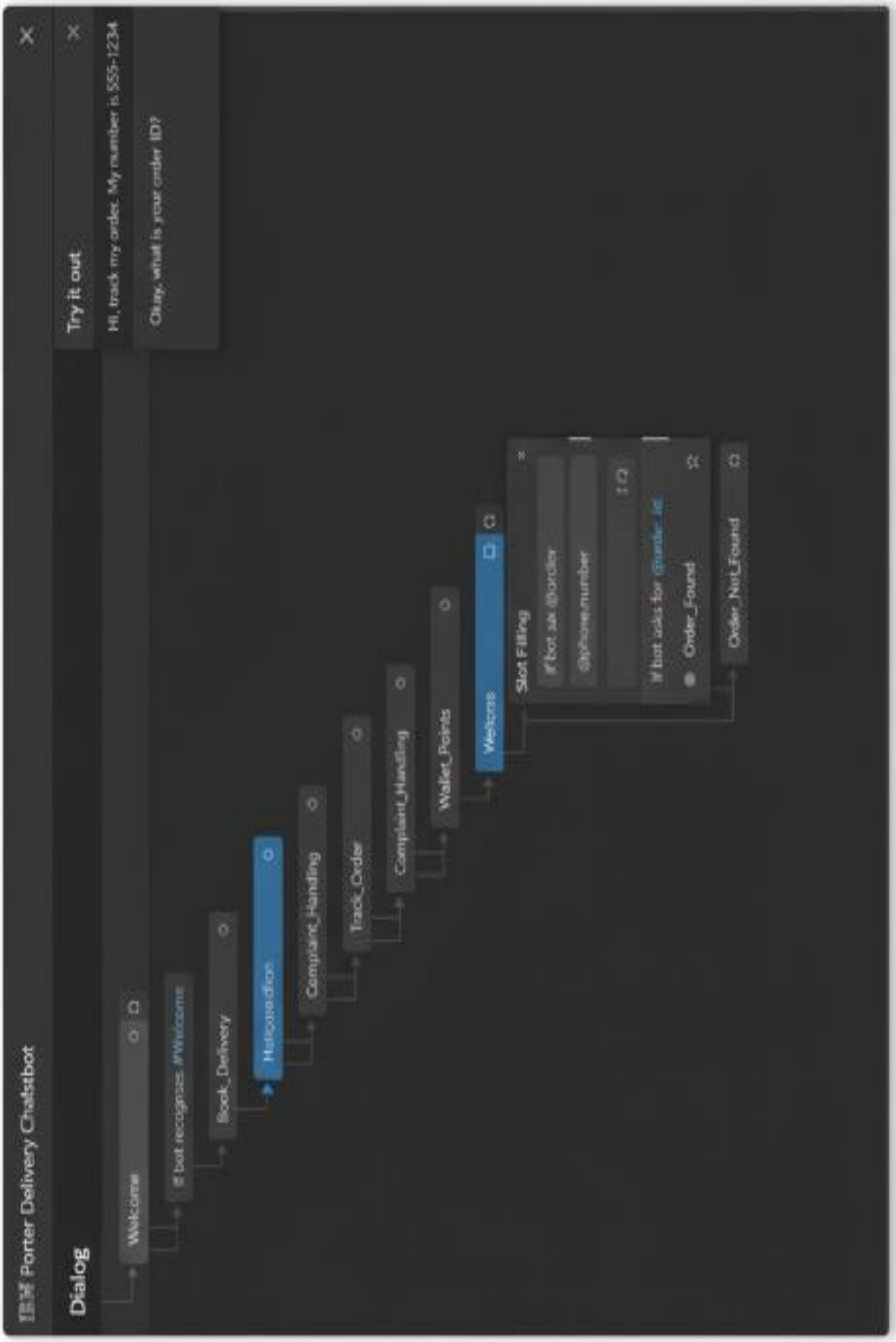
Delivery agent contact number



Appendix B – Screenshots:

IBM Watson Assistant Interfacce

Chatbot Dialog Node Flow



Watson Cloud Deployment Dashboard